discopt: A Differentiable Mixed-Integer Nonlinear Programming Solver with a Memory-Safe Interior Point Method

John R. Kitchin

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1 Abstract

We present discopt, a hybrid Rust/Python solver for mixed-integer nonlinear programming (MINLP) problems, which integrates POUNCE, a separate companion project providing a faithful Rust translation of Ipopt's interior point algorithm, as its default NLP backend. The numerical core is Rust and `numpy`: an ordinary solve imports no JAX, which is an optional dependency of the differentiable-solving, sensitivity, and legacy-evaluator subsystems described below. The system makes six contributions: (1) differentiable MINLP solving via implicit differentiation of the KKT system, enabling decision-focused learning through nonconvex optimization layers; (2) batched node evaluation over the NLP relaxations at branch-and-bound nodes, served by POUNCE's Rust `solve_nlp_batch` after an earlier `jax.vmap` implementation; (3) a JAX-free derivative path built on POUNCE's Rust AD tape, which replaced an in-repo pure-JAX interior point method and agrees with the JAX evaluator to 7.8×10^{-13} on the Lagrangian Hessian and better than 10^{-14} on gradients and Jacobians across 66 corpus instances; (4) integration of POUNCE, a memory-safe Rust IPM with zero C/Fortran dependencies that implements Ipopt's numerical hardening through a source-level translation; (5) a compositional relaxation framework built on a single factorable pass, with McCormick envelopes, rigorous interval-Hessian α BB underestimators, and piecewise McCormick relaxations in an adaptive-partitioning global solver; and (6) a neural network embedding module that formulates trained feedforward networks as algebraic MINLP constraints via full-space, big-M, and reduced-space strategies, enabling optimization over ML surrogates. The linear and mixed-integer subproblems throughout — every spatial-B&B node relaxation and the AMP MILP master — are solved by a self-hosted, hardened pure-Rust revised simplex that retires an external-solver fallback chain in favour of in-engine numerical robustness (iterative refinement, condition monitoring, and EXPAND anti-degeneracy). The global solver couples spatial branch and bound with a cutting-plane framework (reformulation-linearization, second-order-cone, and moment/PSD separators under a structure-gated `cuts`'auto'= policy), complementarity/MPEC reformulations, automatic geometric-program detection with log-space convex routing, and irreducible-infeasible-subsystem diagnosis; differentiability extends to mixed-integer programs via a fix-and-differentiate restriction. Both NLP backends

(POUNCE and Ipopt) produce identical optimal solutions on a suite of 9 NLP and 24 MINLP benchmark problems with zero incorrect results. Because a global solver's product is its *certificate*, we report a systematic soundness audit — roughly three dozen tracked correctness fixes plus a twenty-module code review — that hardened the certificate across the relaxation, presolve, branch-and-bound, cut, parser, and dual-bound layers while maintaining a zero-incorrect-result gate. In a head-to-head against full-license BARON[1] and SCIP[2] over roughly two hundred MINLPLib instances at a matched 60~s budget, discopt records **zero** correctness violations against either solver — no run ever certified a global optimum at the wrong value or returned an incumbent better than the proven optimum; on the orthogonal *coverage* axis (the fraction of instances certified globally optimal within the budget) the two mature solvers lead, each certifying on the order of 180 instances to discopt's $\approx$ 120, a per-node wall-time gap on a serial engine rather than a soundness deficit (Section 17). On a decision-focused learning task, differentiating through the solver achieves 36% regret reduction versus two-stage predict-then-optimize. The algebraic coefficient extraction path provides 17–226$\times$ speedups for LP and QP subproblems. discopt is available under the EPL-2.0 license, and POUNCE is available as a separate project (`https://github.com/jkitchin/pounce`) under the EPL-2.0 license.

2 Introduction

Mixed-integer nonlinear programming (MINLP) is a powerful framework for optimization problems involving both discrete choices and nonlinear relationships, with applications spanning process synthesis, network design, scheduling, and machine learning pipeline optimization.[3,4] The dominant algorithmic paradigm for deterministic global MINLP is spatial branch and bound (sBB), which enumerates a search tree of progressively tighter variable-bound regions, solving a convex relaxation at each node.[5,6,7]

State-of-the-art MINLP solvers—BARON,[1] Couenne,[8] SCIP,[2] and Bonmin[9] are monolithic C, C++, or Fortran codebases that have accumulated decades of numerical hardening. While effective, this architecture creates three gaps in the current solver ecosystem.

Gap 1: No differentiable MINLP. Recent work on differentiable optimization has produced layers for quadratic programs (OptNet[10]), convex programs (cvxpylayers[11]), and decision-focused learning for combinatorial problems.[12,13,14] However, none of these approaches handle nonconvex MINLP. Differentiating through an MINLP solver would enable end-to-end learning of cost parameters from decision outcomes—a capability with broad applications in process optimization and operations research.

Gap 2: No batched or GPU-accelerated branch and bound. In spatial B&B, each node requires solving an NLP relaxation. Modern accelerators (GPUs, TPUs) excel at batched computation, yet existing MINLP solvers process nodes sequentially with sparse linear algebra tuned for single-instance performance. Solving those node relaxations as a batch — through an array-programming framework such as JAX,[15] or, as discopt ultimately settled on, through a batch entry point in the Rust NLP backend — is largely unexplored.

Gap 3: No memory-safe, dependency-free IPM. Ipopt[16] is the workhorse NLP solver used inside most MINLP solvers. It depends on Fortran BLAS/LAPACK, the MUMPS sparse direct solver, and C++ infrastructure totaling hundreds of thousands of lines of

code. Installing cyipopt on a fresh system requires compiling or locating these Fortran dependencies. A Rust[17] reimplementation would provide memory safety guarantees, zero C/Fortran dependencies, and a clean `cargo build` workflow.

This paper describes discopt and its integration with POUNCE, which together address these three gaps. The contributions are:

1. **Differentiable MINLP** via two mechanisms: the envelope theorem (Level 1) for fast objective sensitivity, and implicit differentiation of the KKT system (Level 3) for full primal–dual sensitivity matrices, enabling decision-focused learning through nonconvex solvers.

2. **Batched node evaluation** over the NLP relaxation solves at B&B nodes, with warm-starting from parent solutions. The batch is dispatched to POUNCE's Rust `solve_nlp_batch`; an earlier `jax.vmap` implementation of the same interface is described in Section 5.1 and is no longer the path taken.

3. **A JAX-free derivative path.** An in-repo pure-JAX interior point method — Mehrotra predictor-corrector,[18] Cholesky-based inertia correction, Ipopt-style filter line search,[16] carry-based `jax.lax.while_loop` for JIT cache reuse — served as the reference implementation and was then retired in favour of POUNCE plus a Rust AD tape, which supplies the same derivatives with no framework dependency and no JIT cold start. We report the equivalence measurements that licensed the swap (Sections 5.1 and 17) rather than the algorithm alone, because the negative result is the useful one.

4. **POUNCE integration**: discopt uses POUNCE, a separate companion project that is a source-level C++ → Rust translation of Ipopt's core IPM algorithm, producing a standalone memory-safe NLP solver with a clean `NlpProblem` trait interface, as one of its NLP backends.

5. **Compositional relaxation framework** driven by a single uniform factorable-relaxation pass over the expression DAG, with McCormick envelopes,[19,20] exact composite convex atoms (log-sum-exp, entropy, norm), α BB underestimators,[21,22] piecewise McCormick relaxations[23] inside the AMP solver (97–98% mean-gap reduction on univariate test functions at $k = 8$ uniform partitions), a sound DCP/SUSPECT-style structural convexity certifier[24,25] that routes certified-convex models to a single-solve fast path, and a generalised-(G-)convexity layer that extends the sound-verdict fast path beyond x-space convexity to log-convex and reparameterisation-convex structure.

6. **Neural network embedding** that formulates trained feedforward networks (ReLU, sigmoid, tanh, softplus) as MINLP constraints via big-M, full-space, and reduced-space strategies, with interval arithmetic bound propagation and ONNX model import.[26,27]

7. **Adaptive Multivariate Partitioning**,[28,29] a complementary global MINLP solver that iterates piecewise-McCormick MILP relaxations with adaptive partition refinement, with the soundness guarantee $LB_k \leq f^* \leq UB_k$ at every iteration.

8. **Model-based design of experiments** under `discopt.doe`: D / A / E / ME-optimal designs on a shared `Experiment` interface, batch / parallel MBDoE,[30,31] identifiability and estimability diagnostics including profile likelihood[32] and the Yao orthogonalisation[33] / Brun collinearity index,[34] and a five-criterion model-discrimination loop covering Hunter–Reiner,[35] Buzzi-Ferraris–Forzatti,[36] Jensen–R{\'e}nyi,[37] Lindley mutual information,[38,39] and DT-compound[40] designs.

9. **Multi-objective optimisation** under `discopt.mo`: scalarisation-based Pareto front tracing via weighted sum, AUGMECON2 ε-constraint,[41] weighted Tchebycheff, NBI,[42] and NNC,[43] with hypervolume / IGD / spread / ε-indicator quality metrics[44,45] and full inheritance of the MINLP / differentiable / global-solver stack on every scalarised subproblem.

10. **Cutting-plane framework** with reformulation-linearization (RLT),[46] second-order-cone, and moment/PSD eigenvalue separators,[47] governed by a structure-gated `cuts`'auto'= policy that selects the separator family from the detected problem structure.

11. **Complementarity constraints (MPEC)** under `discopt.mpec`: a `complementarity(f, g)` declaration with three exact reformulations — Scholtes regularisation,[48] SOS1,[49] and generalized-disjunctive lowering[50] — each reducing to a standard discopt `Model`.

12. **Automatic geometric-program detection** under `discopt.gp`: a sound log-convexity verdict[51,52] that recognises posynomial programs and routes them through a single log-space convex solve, extended to *GP-structured MINLP* (integer-augmented posynomial models solved by integer branch-and-bound over log-space convex node relaxations).

13. **Differentiable mixed-integer programming** via fix-and-differentiate, and **infeasibility analysis** (irreducible infeasible subsystems[53] and conflict-driven no-good cuts[54]).

14. **Optimisation under uncertainty**: a stochastic-programming module under `discopt.stochastic` (two-stage recourse and extensive form, multicut L-shaped / Benders decomposition,[55] progressive hedging,[56] sample-average approximation with Mak–Morton–Wood bounds,[57] and CVaR / risk-averse objectives[58,59]) and a bilevel-optimisation module under `discopt.bilevel` that reduces a Stackelberg leader–follower program with a convex lower level to a single-level MPEC via KKT or strong-duality reformulation.[60]

15. **A certificate-soundness audit.** Because a global solver's product is its *certificate*, we report a systematic soundness audit — roughly three dozen correctness issues (all fixed) plus a twenty-module code review — that hardened the certificate across relaxation math, presolve/FBBT, branch-and-bound status logic, cuts, the `.nl` parser, and the LP/dual layer, maintaining =incorrect_count = 0= on the certifying panel (Section 15).

The remainder of this paper is organised as follows. Section 4 describes the discopt architecture, including the modelling API, DAE collocation, neural network embedding, and compiler infrastructure. Section 5 presents the three NLP solver backends. Section 7 details the convex relaxation framework, including the structural convexity certifier of Section 7.6. Section 8 introduces the AMP global solver, and Section 9 the branch-and-bound

search enhancements (primal heuristics, node selection, conflict cuts, and infeasibility analysis). Section 10 covers differentiable optimisation, including differentiable mixed-integer programs. Section 11 discusses dynamic optimisation, Section 12 parameter estimation, Section 13 model-based design of experiments (with identifiability / estimability / discrimination diagnostics), and Section 14 multi-objective optimisation. Section 15 records the certificate-soundness audit. Section 17 reports computational experiments. The final two sections discuss limitations and future work, and conclude.

3 Background and Related Work

3.1 Spatial Branch and Bound for MINLP

Spatial branch and bound (sBB) is the dominant paradigm for deterministic global optimization of MINLP problems.[3] The algorithm maintains a search tree where each node represents a subproblem defined by tightened variable bounds. At each node, a convex relaxation of the original nonconvex problem is solved; the relaxation's optimal value provides a valid lower bound on the global optimum within that region.

The framework was introduced by Land and Doig[5] for integer programming and extended by Dakin.[6] Smith and Pantelides[7] formalized spatial B&B for nonconvex MINLP by combining variable-bound branching with factorable relaxation construction.

Modern sBB solvers enhance the basic framework with bound tightening techniques—feasibility-based bound tightening (FBBT) and optimization-based bound tightening (OBBT)[61,62]—which propagate bound improvements through the expression graph, and primal heuristics such as the feasibility pump[63] that attempt to find good incumbent solutions early in the search.

3.2 Convex Relaxations

The quality of the convex relaxation at each node directly determines the effectiveness of the B&B search. Weaker relaxations produce loose lower bounds, leading to more nodes explored before convergence.

McCormick relaxations[19] construct convex and concave envelopes for factorable functions by composing relaxations of elementary operations (addition, multiplication, univariate functions). Tsoukalas and Mitsos[20] extended this to the multivariate case. The α BB method[21,22] adds a separable quadratic perturbation to ensure convexity, where the perturbation magnitude α is determined by the minimum eigenvalue (or Gershgorin estimate) of the objective Hessian.

Piecewise McCormick relaxations[23] partition the variable domain into subintervals and construct tighter McCormick envelopes on each partition, significantly reducing the relaxation gap at the cost of additional auxiliary variables. Outer approximation (OA)[64,65] and reformulation-linearization technique (RLT)[46] provide complementary cutting-plane approaches.

3.3 Interior Point Methods for NLP

The continuous NLP relaxation at each B&B node is typically solved by an interior point method (IPM). The primal-dual IPM[66,67] transforms inequality constraints into barrier terms in the objective, then applies Newton's method to the resulting KKT system. Key algorithmic components include the Mehrotra predictor-corrector step,[18] which adaptively adjusts the centering parameter for faster convergence, and filter line search,[68,16] which accepts steps that improve either the objective or constraint feasibility.

Ipopt[16] is the most widely used open-source NLP solver, implementing a filter line-search interior point method with inertia correction, feasibility restoration, and warm-starting. It has accumulated over 20 years of numerical hardening for edge cases including degenerate Jacobians, rank-deficient KKT systems, and cycling detection.

Sensitivity analysis of NLP solutions with respect to problem parameters is well-established.[69] Pirnay et al.[70] implemented the sIPOPT framework within Ipopt for efficient parametric sensitivity computation by reusing the KKT factorization.

3.4 Differentiable Optimization: Prior Work

The idea of differentiating through an optimization solver to enable end-to-end learning has received significant recent attention. Amos and Kolter[10] introduced OptNet for differentiating through quadratic programs. Agrawal et al.[11] extended this to general convex programs via cvxpylayers. Elmachtoub and Grigas[12] formalized the "smart predict, then optimize" framework, and Wilder et al.[13] and Mandi et al.[14] developed methods for combinatorial optimization with decision-focused learning.

A critical gap remains: **none of these approaches handle nonconvex MINLP**. Existing differentiable optimization layers require convexity (OptNet, cvxpylayers) or use surrogate loss functions that approximate but do not exactly differentiate through the solver (decision-focused learning for combinatorics). discopt fills this gap by providing exact gradients through a nonconvex NLP solver via implicit differentiation of the KKT conditions.

3.5 Gaps in the State of the Art

Table 1 summarizes the capabilities of existing systems versus discopt.

4 The discopt Architecture

4.1 System Overview

discopt is organized as a three-layer system: (1) a Python modeling API for problem formulation, (2) a `numpy` relaxation and evaluation layer, and (3) a Rust backend for tree management, expression structure detection, presolve, the simplex, and automatic differentiation. A default solve touches only these three; the JAX-backed subsystems of Sections 10 and 7 sit beside the default path rather than under it, and `jax` is not imported unless one of them is used.

Feature	BARON	SCIP	Bonmin	cvxpylayers	OMLT	discopt
Nonconvex MINLP	Yes	Yes	Yes	No	Via Pyomo	Yes
Differentiable solve	No	No	No	Yes (convex)	No	Yes
Batched node evaluation	No	No	No	N/A	No	Yes
Memory-safe solver	No	No	No	N/A	No	Yes
Zero C/Fortran deps	No	No	No	Yes	No	Yes
DFL through MINLP	No	No	No	No	No	Yes
Compositional relaxation	Yes	Partial	No	N/A	No	Yes
NN embedding in MINLP	No	No	No	No	Yes	Yes
Robust counterpart API	No	No	No	No	No	Yes
Adjustable RO (ADR)	No	No	No	No	No	Yes

Table 1: Comparison of discopt with existing optimization frameworks. DFL = decision-focused learning.

```
User Model (Python)
     |
     v
Expression DAG ---> uniform factorable pass ---> numpy relaxation
     |                                             |
     v                                             v
Rust Expression IR             AD tape (obj, grad, Hess, Jac)
     |                                             |
     v                                             v
B&B Tree Manager               NLP Solver Backend
(node pool, branching,         (POUNCE / Ipopt)
 bound tightening,                     |
 revised simplex)                      v
     |                                 |
     +-------- batch dispatch --------+
                    |
                    v
              SolveResult

(optional, off the default path)
Expression DAG ---> JAX DAG compiler ---> jit / grad / vmap
                          |
                          v
     differentiable solve, parametric sensitivity,
     model-based DoE, legacy-evaluator fallback
```

An automatic problem classifier inspects the expression DAG to categorize problems as LP, QP, NLP, MILP, MIQP, or MINLP. For LP and QP subclasses, discopt dispatches to specialized solvers that extract algebraic coefficients directly from the DAG, bypassing expression-level autodiff entirely. This algebraic extraction path provides 17–226× speedups over the generic NLP path (\S 7.4).

Core and plugins. To keep the core lean, the domain-specific *application builders* — model-based design of experiments (`discopt.doe`), the teaching / tutor front-ends, and the alternating-current optimal-power-flow and pooling builders (`discopt.opf`, `discopt.pooling`) — have been extracted into standalone `discopt-doe`, `discopt-course`, and `discopt-apps` distributions. Because `discopt` is a namespace package, installing a plugin restores its `discopt.<name>` import path unchanged, and each plugin registers its command-line front-

end through a `discopt.cli` entry-point hook so that `discopt <subcommand>` dispatches to it. A stable public `discopt.parametric` API compiles model expressions into JAX-differentiable functions of (x, p) without reaching into internal modules, giving both in-tree consumers (parameter estimation) and external plugins (DoE, microkinetics) a single supported contract. The core retains the general MINLP machinery — GDP, DAE collocation, neural-network embedding, robust and stochastic optimisation — while domain builders live as plugins.

4.2 Modeling API

The modeling API uses operator overloading to construct an expression DAG that maps to Rust expression IR, to the `numpy` relaxation pass, and — for the optional differentiable subsystems — to JAX-traceable functions:

```python
import discopt.modeling as dm

m = dm.Model("portfolio")
x = m.continuous("weights", shape=(n,), lb=0, ub=1)
y = m.binary("selected", shape=(n,))

m.minimize(risk @ (x * x) - expected_return @ x)
m.subject_to(dm.sum(x) == 1, name="budget")
m.subject_to(x <= y, name="indicator")  # x_i > 0 => y_i = 1
m.subject_to(dm.sum(y) <= k, name="cardinality")

result = m.solve()
```

The API supports continuous, binary, and integer variables; parameters (for sensitivity analysis and decision-focused learning); generalized disjunctive programming (GDP) constraints with big-M, multiple big-M (MBM) with LP-based bound tightening, convex hull, and Logic-based Outer Approximation (LOA[71]) reformulations via `gdp_method`; `BooleanVar` with propositional logic operators (`&`, `|`, `~`, `.implies()`, `.equivalent_to()`) and CNF-to-linear conversion; nested disjunctions for hierarchical process design; and standard mathematical functions (`exp`, `log`, `sin`, `cos`, `sqrt`, `abs`, `tanh`, `sigmoid`, `softplus`, and matrix operations).

4.3 Dynamic Optimization via DAE Collocation

The `discopt.dae` module extends the modeling API to dynamic optimization problems by transcribing ODE/DAE systems into algebraic NLP constraints using orthogonal collocation on finite elements.[72,73] Users declare state, algebraic, and control variables, supply an ODE/DAE right-hand side as a Python callable that builds discopt expressions, and call `discretize()` to generate all collocation and continuity constraints automatically. Radau IIA and Gauss-Legendre schemes with 1–5 points per element, second-order ODEs via state augmentation, index-1 DAEs, piecewise-constant controls, and integral objectives are all supported. The transcription is covered in detail in the Dynamic Optimization section (11); parameter estimation and model-based design of experiments for dynamic systems are built on the same machinery (12, 13).

4.4 Neural Network Embedding

The `discopt.ml` module embeds trained feedforward neural networks as algebraic constraints in MINLP models, enabling optimization over machine learning surrogates with global optimality guarantees. This follows the approach of OMLT,[26] adapted to discopt's expression DAG, automatic differentiation, and McCormick relaxation framework.

An `NNFormulation` builder (following the same pattern as `DAEBuilder`) takes an existing `Model`, a `NetworkDefinition` describing the trained network's weights, biases, and activation functions, and a formulation strategy, then adds all necessary variables and constraints:

```
from discopt.ml import NNFormulation, NetworkDefinition, DenseLayer, Activation

net = NetworkDefinition([
    DenseLayer(W1, b1, Activation.RELU),
    DenseLayer(W2, b2, Activation.LINEAR),
], input_bounds=(lb, ub))

nn = NNFormulation(model, net, strategy="relu_bigm")
nn.formulate()
model.minimize(dm.sum(nn.outputs))
```

Three formulation strategies are provided:

- **Full-space**: Explicit pre-activation ($\hat{z}$) and post-activation (z) variables for each neuron, with affine constraints $\hat{z} = Wx + b$ and activation constraints $z = \sigma(\hat{z})$. Supports smooth activations (sigmoid, tanh, softplus) that integrate directly with the AD tape and McCormick relaxations for spatial B&B.
- **ReLU big-M**.[27] For networks with ReLU activations, introduces binary variables q_j and linear big-M constraints: $z \geq 0$, $z \geq \hat{z}$, $z \leq u \cdot q$, $z \leq \hat{z} - l(1-q)$, where l, u are pre-activation bounds from interval arithmetic. Neurons with bounds that are always positive (or always negative) are detected and simplified to avoid unnecessary binary variables.
- **Reduced-space**: Builds a single nested expression with no intermediate variables, suitable for small networks.

Interval arithmetic bound propagation through the network layers computes tight big-M constants by splitting weight matrices into positive and negative parts. An ONNX reader (`load_onnx`) imports trained models from scikit-learn, PyTorch, and Keras via the ONNX interchange format. Beyond feedforward networks, the module also embeds decision trees and tree ensembles via a per-leaf MILP encoding.

The soundness audit of Section 15 closed two embedding-specific hazards. For a network embedded behind an affine input/output scaling, bound propagation now runs over the *scaled* box the layers actually consume, so the ReLU big-M constants are a valid overestimate rather than being computed from the wrong (unscaled) domain; and the per-leaf big-M in the tree-ensemble encoding is now a per-constraint constant that stays inert for a non-selected leaf even when a split threshold falls outside the variable box, so an out-of-box threshold no longer cuts feasible points. On the reader side, the ONNX path now applies

`Gemm` scaling attributes, consumes residual `Add` biases correctly, and rejects non-sequential (branched) graphs instead of silently flattening them, while the scikit-learn reader respects the output activation of an MLP classifier and the base score of a gradient-boosted ensemble.

4.5 Trainable Surrogates: The Other Regime

Everything above is the *frozen* regime: the weights are constants, and the optimization runs over a fixed network. `discopt.ml.trainable` is the complementary regime — the weights are decision `Variable` objects, and the surrogate is *fitted* inside the same NLP that solves the physics. This is the simultaneous approach to training neural differential-algebraic systems of Lueg et al.,[74] in which a neural rate law inside a collocation DAE is trained jointly with the dynamics rather than by an outer loop around a simulator (contrast the adjoint-through-an-integrator formulation of neural ODEs[75]).

Three design invariants make this composable rather than a second framework. A surrogate emits **symbolic** expressions built from discopt's own intrinsics — never an opaque Python callable, which would disable global certification, integer variables and `.nl` export. Layers are emitted in **matrix** form, with array-shaped weight variables and `@`, not per-neuron scalar loops. And activations are **smooth only**: `TrainableDense` accepts linear, tanh, sigmoid and softplus, and refuses ReLU with a message naming softplus as the smooth stand-in. The refusal is a design decision with a stated reason rather than a measured result: on the frozen path a ReLU kink is handled by binaries and big-M, which is available precisely because the weights are constants, whereas on the trainable path the kink would have to be resolved inside the training NLP — either a nonsmooth gradient under an interior-point method or a combinatorial layer over trainable weights.

`TrainableNetwork` stacks `TrainableDense` layers and supplies a deterministic Glorot-style warm start, $W \sim \mathcal{N}(0, (s/\sqrt{n_{\text{in}}})^2)$ clipped to the weight box. `TrainableKernelExpansion` emits $r(x) = \sum_j \alpha_j \exp(-(x - c_j)^2/2\ell^2)$ with the centres and lengthscale fixed by design, so the block is *linear in its parameters* and contributes no nonconvexity of its own — zero initialisation suffices and there are no weight-symmetry local minima, the best-conditioned of the trainable surrogates. Its hyperparameters are deliberately kept outside the NLP, because the marginal-likelihood objective that would tune them needs matrix inverses and determinants that do not fit an algebraic NLP cleanly.

The framework is open at this boundary rather than closed around these three classes. A `Surrogate` protocol declares the duck-typed contract — `__call__(x)` returning an expression, plus `parameters`, `n_parameters`, `l2_penalty` and `initial_values` — of which only `__call__` is load-bearing; conformance is structural, so nothing need inherit from it. The solver is entirely decoupled: the surrogate participates only by being called inside the user's right-hand side, expression in and expression out, and neither `train` nor `fit_trajectories` calls it directly. The real admissibility criterion is mathematical, not structural: **a surrogate is trainable in this NLP exactly when it reduces to a smooth expression in continuous decision variables.** Neural nets with smooth activations, Gaussian-process and kernel means, polynomials, splines, differentiable decision trees, and any fixed-structure symbolic formula whose constants are fitted all qualify; non-smooth or discrete structure — ReLU, hard-split trees, or *searching* a symbolic structure — belongs on the frozen path or in an MINLP, not in the smooth simultaneous NLP.

Multi-experiment fitting is composition over existing primitives, with no new solver mathematics. `dae.fit_trajectories` builds one collocation transcription per experimental trajectory on a *shared* model and wires every block to the *same* right-hand-side callable. Because that callable closes over the surrogate's weight variables, every block emits constraints referencing the same weights, and they are trained jointly in one sparse NLP; the per-trajectory initial conditions come from the trajectory, and states absent from a trajectory's data are simply unobserved. `TrajectoryFit` supplies the weighted least-squares objective (using exact collocation-polynomial interpolation at the measurement times), a data-interpolated warm start for the state variables that merges with the surrogate's own initial values, and an extractor that reads a fitted trajectory back out.

`train()` is deliberately thin, and its docstring states a **negative** guarantee: it validates the warm start, builds an evaluator — with a Gauss–Newton Hessian approximation when the objective is recognised as a sum of squares, falling back to the exact Hessian automatically when it is not — and calls the local NLP backend directly. It does **not** route through `Model.solve`, so the returned result carries **no global optimality certificate**. That is the honest description of the regime: simultaneous training is a large structured local NLP. The bridge to the certified path runs in both directions — `freeze(values)` extracts trained weights into a frozen `NetworkDefinition` that the formulations of the preceding section then embed and optimize over globally, and `TrainableNetwork.from_definition` turns any of the existing ONNX, scikit-learn and PyTorch readers into *initialisers* rather than only freezers, with no reader change.

The design rests on measurements taken during the exploration that preceded it, all recorded in-tree: a neural-DAE prototype converging in 19 interior-point iterations and 1.3 s at 19 weights; full-space training of 1021 weights across 3 trajectories in 17 iterations and 37 s from a Glorot start; the Gauss–Newton detector firing on collocation least-squares sums and cutting evaluator build time from 0.78 s to 0.02 s; and the kernel surrogate converging in 12 iterations from zero initialisation with no restarts. The 1021-weight result is the one that mattered most, because it *falsified* the plan it belonged to: the premise that paper-scale training would require decomposition machinery was abandoned on the strength of it, and none was built. These figures date from the exploration and are gated against regression by the test suite rather than re-measured for this paper.

4.6 Robust Optimization Module

The `discopt.ro` module converts a nominal model with uncertain parameters into a deterministic **robust counterpart** that is feasible for every realization of the uncertainty within a prescribed set. Following the classical framework of Ben-Tal and Nemirovski,[76,77] robust optimization replaces the original problem

```
min_x  f(x, p)     s.t. g_i(x, p) <= 0  for all p in U
```

with a deterministic equivalent that encodes the worst-case constraint tightening analytically.

4.6.1 Uncertainty Sets

Three standard uncertainty set families are supported:

- **Box** (L_∞ ball) $\mathcal{U} = \{p : |p_j - \bar{p}_j| \leq \delta_j\}$: the worst-case maximisation of an affine function $c^\top p$ over the box is $\bar{c}^\top p + \sum_j \delta_j |c_j|$, yielding a separable 1-norm penalty. When p appears only as an additive RHS term, this collapses to a single bound substitution $p \leftarrow \bar{p} \pm \delta$.
- **Ellipsoidal** $\mathcal{U} = \{p : \|\Sigma^{-1/2}(p - \bar{p})\|_2 \leq \rho\}$: the worst-case is $\bar{a}^\top p + \rho \|\Sigma^{1/2} a(x)\|_2$, a second-order cone (SOC) penalty.[76]
- **Polyhedral** $\mathcal{U} = \{p : A\xi \leq b,\ \xi = p - \bar{p}\}$: the support function is evaluated by LP duality, introducing auxiliary dual variables $\lambda \geq 0$ satisfying $A^\top \lambda = a(x)$ and adding $b^\top \lambda$ to the constraint.[78]

The **budget-of-uncertainty** set (Bertsimas–Sim[78]) is provided as a convenience constructor:

$$\mathcal{U}_\Gamma = \left\{ p : |p_j - \bar{p}_j| \leq \delta_j,\ \sum_j \frac{|p_j - \bar{p}_j|}{\delta_j} \leq \Gamma \right\}$$

The budget $\Gamma \in [0, k]$ interpolates between the nominal solution ($\Gamma = 0$) and full box robustness ($\Gamma = k$), controlling the trade-off between solution cost and protection level.

4.6.2 Implementation: Reformulation Strategies

The box reformulation uses two strategies depending on the structure of the expression DAG. When uncertain parameters appear with constant (known-sign) coefficients, the reformulation traverses the DAG while tracking the **effective sign** of each uncertain parameter at each node. At a subtraction node $a - p$, the parameter p acquires sign -1; at a negation node $-p$, the sign flips. Multiplication by a non-negative constant preserves the sign; multiplication by a negative constant flips it. This sign determines whether the worst-case value is the upper or lower component bound:

```
sign > 0 and maximize  =>  use upper bound  (pbar + delta)
sign < 0 and maximize  =>  use lower bound  (pbar - delta)
```

When the expression contains **bilinear** terms ($p \cdot x$ where both are free), as arises from affine decision rules, the coefficient of p involves decision variables whose sign is not known a priori. In this case the reformulation extracts the per-component coefficient of each uncertain parameter and applies absolute-value linearization: the worst-case of $\text{coeff}(x) \cdot \xi$ over $|\xi| \leq \delta$ is $\delta \cdot |\text{coeff}(x)|$, where the absolute value is linearized via auxiliary variables $t \geq 0$ satisfying $\text{coeff}(x) \leq t$ and $-\text{coeff}(x) \leq t$. The implementation automatically detects the presence of bilinear terms and selects the appropriate strategy.

For the polyhedral uncertainty set, the reformulation introduces auxiliary dual variables $\lambda \geq 0$ (one per polytope row) with equality constraints $A^\top \lambda = \text{coeff}(x)$ and a worst-case

penalty $b^\top \lambda$. This LP-duality approach correctly handles the budget-of-uncertainty constraint, where the budget Γ couples the uncertainty components and component-wise bounds alone would be insufficient.

The `RobustCounterpart` class follows the same builder pattern as `NNFormulation`:

```
from discopt.ro import BoxUncertaintySet, RobustCounterpart

m = dm.Model("production")
x = m.continuous("x", shape=(3,), lb=0)
c = m.parameter("c", value=[10.0, 15.0, 8.0])
d = m.parameter("d", value=100.0)

m.minimize(dm.sum(c * x))
m.subject_to(dm.sum(x) >= d, name="demand")

rc = RobustCounterpart(m, [
    BoxUncertaintySet(c, delta=0.10 * c.value),   # costs ś 10%
    BoxUncertaintySet(d, delta=0.05 * d.value),   # demand ś 5%
])
rc.formulate()          # modifies m in-place; m is now deterministic
result = m.solve()
```

After `formulate()`, the model contains no uncertain parameters: each has been replaced by its worst-case constant. The reformulated model is a standard MINLP that can be solved by any of discopt's solver backends.

4.6.3 Correctness Guarantee

For the additive/affine uncertainty case, both reformulation strategies produce an exact robust counterpart: the reformulated constraints hold for all $p \in \mathcal{U}$ if and only if they hold after substitution. The sign-tracking path follows directly from LP duality;[76] the coefficient-extraction path is exact because the absolute-value linearization is tight for box uncertainty. The polyhedral reformulation via LP duality is exact for any polyhedral set, including the budget-of-uncertainty case.[78] The ellipsoidal reformulation produces exact second-order cone constraints for affine uncertainty. The implementation verifies correctness via structural tests that confirm (1) no uncertain parameters remain in the reformulated DAG, and (2) the reformulated model solves to the analytically derived robust optimal value.

The soundness audit of Section 15 sharpened this guarantee in three places. First, a bare parameter–variable product $p \cdot x$ with a sign-*indefinite* variable factor is now routed through the absolute-value linearization rather than the sign-tracking fast path (which is sound only when the variable is provably non-negative), closing an under-protection hole. Second, `RobustCounterpart.formulate()` now asserts that *no* uncertain parameter survives robustification, raising rather than silently returning a non-robust model when a pattern is unsupported. Third, the budget-of-uncertainty constructor was corrected to encode the *exact* Bertsimas–Sim set:[78] the earlier implementation stored only the all-plus/all-minus budget facets and so protected against a strict superset of the true set — sound (over-protective by up to a factor of two in mixed-sign directions) but more conservative than the promised set. The corrected counterpart stores the compact lifted (ξ, u) representation, which the polyhedral LP-duality path dualises exactly; it was verified robust by exhaustive in-set sampling (worst in-set constraint slack within tolerance across all vertices, a dense grid, and interior samples) while recovering the less-conservative Bertsimas–Sim optimum.

4.6.4 Adjustable Robust Optimization via Affine Decision Rules

Static robust optimization requires all decisions to be fixed before the uncertain parameters are revealed. When some decisions can be deferred until after (partial) observation of the uncertainty, adjustable robust optimization (ARO)[79] provides a less conservative solution.

The two-stage framework partitions variables into **here-and-now** decisions x (chosen before observing ξ) and **wait-and-see** recourse variables $y(\xi)$ (chosen after). Optimising over arbitrary functions $y(\cdot)$ is generally intractable (the problem is semi-infinite), so discopt adopts ***affine decision rules*** (also called linear decision rules).[79,80]

$$y(\xi) = y_0 + \sum_{j=1}^{n_\xi} Y_j \cdot \xi_j, \quad \xi_j = p_j - \bar{p}_j$$

where y_0 and $\{Y_j\}$ are new ordinary decision variables representing the intercept and policy matrix columns. Bertsimas and Goyal[81] establish that for LP problems with right-hand-side uncertainty, affine rules are **optimal**, meaning no other measurable recourse policy achieves a lower cost; for uncertain constraint matrices they are an inner approximation.

The `AffineDecisionRule` class implements this substitution as a pre-processing step compatible with the existing `RobustCounterpart`:

```
from discopt.ro import AffineDecisionRule, BoxUncertaintySet, RobustCounterpart

# Stage 1: substitute recourse variable y with y0 + Y0 * (d - dbar)
adr = AffineDecisionRule(y, uncertain_params=d)
adr.apply()          # retires y; adds y0, Y0 to model._variables

# Stage 2: static robust counterpart handles remaining xi terms
rc = RobustCounterpart(m, BoxUncertaintySet(d, delta=20.0))
rc.formulate()       # fully deterministic; optimise over (x, y0, Y0)
```

The DAG substitution walks every constraint and objective node, replacing Variable nodes that match the recourse variable with the affine expression via identity comparison ($O(|V| + |E|)$ in the DAG size). After substitution the recourse variable is retired from the model's variable list and remaining variables are renumbered. In addition, `apply()` automatically enforces robust feasibility of the recourse variable bounds: if the original variable y had bounds $[l, u]$, the constraints $y_0 + \sum_j Y_j \xi_j \leq u$ and $y_0 + \sum_j Y_j \xi_j \geq l$ are added to the model, ensuring that the affine rule respects the original domain for all realizations of ξ. The subsequent `RobustCounterpart.formulate()` call handles the bilinear $Y_j \cdot \xi_j$ terms in these bound constraints via coefficient extraction and absolute-value linearization.

A call to `AffineDecisionRule.apply()` followed by `RobustCounterpart.formulate()` leaves the model fully deterministic with (1) no recourse variables, (2) no uncertain parameters, and (3) new intercept and policy-column variables in their place. For simple problems (e.g., a two-variable LP with a single uncertain parameter), the affine decision rule typically matches the static robust optimal value, because the static solution already adapts optimally. The theoretical advantage of ADR emerges in problems with structural asymmetry: multiple constraints, integer first-stage variables, or high-dimensional uncertainty where the static worst case is overly conservative.

4.7 Complementarity Constraints and MPEC

Mathematical programs with equilibrium (complementarity) constraints (MPECs)[50] arise whenever an inner equilibrium — contact mechanics, Karush–Kuhn–Tucker conditions of a lower-level problem, piecewise-linear device characteristics, or economic complementary slackness — is embedded in an outer optimisation. The defining relation is the complementarity condition $0 \le f(x) \perp g(x) \ge 0$, requiring $f, g \ge 0$ component-wise together with $f_i\, g_i = 0$. The product constraint violates every standard constraint qualification at the origin, so a naive nonlinear formulation stalls at spurious stationary points.

The `discopt.mpec` module exposes a single declarative primitive, `complementarity(f, g, name...)`=, and lowers a set of such conditions to a standard discopt `Model` through three reformulations selected by `solve_mpec(model, pairs, method...)`=. Two of the three *encode* the condition exactly; the third *relaxes* it, and the distinction is load-bearing rather than terminological:

- **Scholtes regularisation** (`method`"scholtes"=, the default). The exact product $f_i\, g_i = 0$ is relaxed to $f_i\, g_i \le t$ and a homotopy drives $t \downarrow 0$ through a short sequence of local NLP solves.[48] Scholtes established that limit points of this scheme are C-stationary, and B-stationary under mild conditions, making it the method of choice when a fast locally-optimal MPEC solution suffices. Its rows admit points the relation forbids, so a solve over them certifies the *relaxation* and not the declared model; `mpec.RELAXING_METHODS` records exactly that, and the arm reports a `local_*` status carrying no dual bound.

- **SOS1** (`method`"sos1"=). The pair (f_i, g_i) is placed in a special-ordered-set-of-type-1,[49] encoding "at most one of f_i, g_i is nonzero" exactly; branch and bound then resolves the discrete choice, yielding a globally valid solution.

- **Disjunctive lowering** (`method`"gdp"=). The condition is expressed as the disjunction $(f_i = 0) \vee (g_i = 0)$ and handed to the existing GDP machinery (Section 4), which applies big-M or convex-hull reformulation with the same bound-tightening used elsewhere.

Because all three reformulations emit an ordinary `Model`, an MPEC inherits the full solver stack: the regularised path uses any NLP backend, while the SOS1 and disjunctive paths are solved to global optimality by spatial branch and bound with the relaxations of Section 7. Nonlinear complementarity operands f, g are lifted to auxiliary variables so the disjunctive and big-M encodings remain exact on bilinear and higher-degree expressions.

4.7.1 A complementarity relation is model state, and the residual has to be measured against the declaration

The hazard that dominates this module is not the lowering algebra; it is a residual measured against the wrong rows, printing a small number, and being believed. A complementarity relation lives *outside* `_constraints` — it is a declaration the solver lowers into rows — and it was originally three fields that every model-rebuilding pass discarded. Measured before the fix, the relation count went `before=1, after=0` through GDP lowering under each of `big-m`, `hull` and `mbigm`, through integer-product expansion, and through factorable reformulation

alike; all that survived a lowering was the pair name baked into generated identifiers. A source residual computed on such a model would have measured the *relaxed row* rather than the source product.

The relation is therefore now a first-class object carrying its **source** operands — never rewritten by any pass, so a residual computed from them is a residual of the declared condition — together with the bounds the relation itself declares on each operand, an optional characteristic scale, a provenance `role` (`ncp_pair`, `box_mcp`, `from_kkt`), the parent construct that generated it, and, for an element of a vector relation, its multi-index and a back-pointer to the declared parent. Every rebuilding pass forwards the set explicitly through `carry_complementarities`, which raises — naming both the relation and the pass — rather than dropping one whose provenance it cannot resolve. Two consequences are worth stating because they were each learned from a defect. The role is provenance and never a lowering switch: every lowering branches on the relation's *declared operand bounds*, because `Complementarity` is publicly constructible and a hand-built relation with $g \in [-1, 1]$ carrying the default role once reached the lowerings and emitted $g \geq 0$ against a declared lower bound of -1. And whether a relation has been lowered is a fact about the *model*, not about the relation: a relation is a shared object, so the mark and the method that produced it live on the model in an identity map, after an earlier design that stored weak references on the relation made a `Model` unpicklable and silently lost the mark under `deepcopy`, and a later one that left the *method* on the relation reported `sos1` for a model carrying disjunctive rows.

Three false certificates, all measured, fix the shape of the guards that follow. (i) On the balanced probe $\max x + y$ s.t. $x = y$, $0 \leq x \perp y \geq 0$ with true optimum 0, a Scholtes run left its relations marked lowered, and a subsequent `solve()` returned `status`'optimal', `objective $2.83 \times 10^{-4}$, =gap_certified=True` — a certified optimum strictly better than the truth, admitted by a guard whose entire purpose was to catch a declared relation solved as if absent. "Has a lowering" was never the right test; "has an **exact** lowering" is. (ii) Because `Model.complementarity` lowers its pair eagerly, a relation handed afterwards to the homotopy was *skipped*: the $fg \leq t$ row was never emitted, nothing in the model depended on t, and the continuation ran ten stages each recorded `optimal`, `accepted`, `certified` — because each subsolve genuinely did converge, on the unregularised problem — and returned a point violating complementarity by 2.0. On $\min(x_0 - 2)^2 + (x_1 - 1)^2$ s.t. $0 \leq x_0 \perp x_1 \geq 0$ over $[0, 3]^2$, whose optimum is 1.0 at $(2, 0)$, the eagerly-lowered pair reported objective 1.67×10^{-9} with $x_0 x_1 = 2.0$ against the pure factory's 1.0 with $x_0 x_1 = 2 \times 10^{-8}$: not an approximation of the right answer but the answer to a different problem, reported as converged. (iii) The gate that promotes a local MPEC point to an incumbent once accepted a *cached* residual report, which records neither a point nor a model; a report taken at the exactly complementary origin admitted $(10^{-4}, 10^{-4})$, whose objective -2×10^{-4} lies below the true optimum — an invalid incumbent cutoff that, fed to a global solve, fathoms the optimum away. Each is closed by a refusal rather than a tolerance: `require_all_relations_lowered` runs at *every* solver entry point (it first lived only in `Model.solve`, and the four in-tree callers of `solve_model` went straight past it), distinguishes relaxing from exact lowerings, and the incumbent gate recomputes the report at the acceptance boundary against the actual candidate, with the `report=` keyword removed rather than left available.

4.7.2 Reporting the residual of the model the user wrote

POUNCE's internal barrier complementarity is a property of the *generated* NLP; the MPCC condition $0 \le f \perp g \ge 0$ is a property of the *user's* model. They are different numbers, and at a Scholtes point they differ by orders of magnitude: on the probe above at $t = 10^{-8}$ the generated rows are violated by 1.0×10^{-8} while the source complementarity $\min(x, y)$ is 1.4×10^{-4}. `discopt.mpec_report` measures the *declared* operands and reports the complementarity residual (the min form, the MCP normal map for a box relation, the product, or Fischer–Burmeister on request), the declared-bound violation per operand, source primal feasibility against the original rows, the binary-selector integrality $\max_i y_i(1 - y_i)$, and the lowered rows' own residual beside the source one so the two can be compared. Each residual carries the worst case its regularisation *admits* — $\sqrt{t}$ for the min form, t for the product, $(2-\sqrt{2})\sqrt{t}$ for Fischer–Burmeister — documented as an upper bound on the admitted residual rather than a limit on attainable accuracy, since $fg \le t$ also admits $(0, 1)$, whose residual is exactly zero at every positive t.

The statuses follow the same logic. `discopt.status` carries `local_optimal`, `local_limit` and `local_infeasible` as statuses rather than as a flag beside `optimal`, because the benchmark harness scores =status == OPTIMAL= with no gap as a *proved* optimum and every consumer that pattern-matches on the status would otherwise inherit the bug. A stalled continuation reports `local_infeasible`, never `infeasible`, which is a certificate in the other direction; `SolveResult` raises if a local status carries a bound or a root bound and forces `gap_certified=False`. Stationarity classifications are never claimed, because discopt checks no C-, M- or S-stationarity conditions.

4.7.3 Mixed complementarity over a box

`Model.mcp(residual, variable, lb`..., ub=...)= states the mixed form $F(z) \perp z \in [l, u]$, which pairs a residual with a *bounded* variable rather than with a second nonnegative expression, and whose condition has three branches: $z = l \Rightarrow F \ge 0$, $l < z < u \Rightarrow F = 0$, $z = u \Rightarrow F \le 0$. The case $l = 0$, $u = +\infty$ *is* `Model.complementarity`, and is recorded and lowered as that pair so the two forms cannot diverge on the case they share. The bounds are never guessed from anything but a plain scalar variable's own declaration, since the bounds *are* the relation's semantics. The general box form is represented and deliberately **not** lowered: handing a three-branch condition to a two-branch lowering would encode a different feasible set and certify it, which is not hypothetical — with $z \in [-1, 1]$, $\max z$ and `mcp(z + 1, z)`, the solver returned $z = 1$, where the $z = u$ branch requires $F \le 0$ and $F = +2$. Such a model is refused at the solver boundary, by a message naming the relations and the two ways to state the problem instead.

4.8 Stochastic Programming

Where the robust-optimisation module of Section 4 protects against *every* realisation in an uncertainty set, the `discopt.stochastic` module takes the complementary, distributional view: a two-stage program with recourse minimises a first-stage *here-and-now* cost plus the expected (or risk-adjusted) cost of a second-stage *wait-and-see* recourse taken once the uncertainty is realised.[59] It is the distributional sibling of `discopt.ro`.

Uncertainty is carried by a `ScenarioSet` — a finite, probability-weighted collection of `Scenario` objects, built explicitly, from a list, or by Monte-Carlo sampling of user-supplied per-parameter samplers (`ScenarioSet.sample`). `build_extensive_form` assembles the deterministic-equivalent *extensive form*: given a first-stage cost and a `recourse_builder(model, scenario_data, s)` callback that stamps the per-scenario recourse constraints, it forms the scalarised objective $c^\top x + \sum_s p_s\, Q_s(x)$ and hands back an ordinary `Model` that the full MINLP stack then solves.

The risk functional is pluggable through a `RiskMeasure` interface. `Expectation` gives the risk-neutral $\sum_s p_s Q_s$; `CVaR($\alpha$)` adds the Rockafellar–Uryasev conditional value-at-risk[58] via its convex epigraph reformulation; `MeanCVaR($\lambda,\alpha$)` interpolates between the two; and `chance_constraint` enforces a sample-average probabilistic constraint $\mathbb{P}(g(x,\xi) \le 0) \ge 1 - \varepsilon$ through per-scenario big-M indicators.

Two decomposition solvers avoid materialising the full extensive form on large scenario trees. `solve_lshaped` runs a probability-weighted multicut L-shaped (Benders) decomposition,[55] dispatching between classical L-shaped LP recourse and generalized Benders for convex-nonlinear recourse. `progressive_hedging` implements the Rockafellar–Wets scenario-decomposition scheme,[56] driving per-scenario first-stage copies to a common non-anticipative value through an augmented-Lagrangian penalty. For solution-quality certification, `saa_lower_bound_estimate` / `saa_upper_bound_estimate` form the Mak–Morton–Wood statistical bounds that bracket the true optimum from independent sample batches,[57] and `optimality_gap_estimate` reports the resulting gap. A distributionally-robust primitive (`worst_case_expected_cost`) evaluates the worst-case expectation over a total-variation (L_1) ambiguity ball around the nominal scenario probabilities; the full single-level DRO reformulation and multistage (nested-Benders / SDDP) recourse are reserved for future work.

4.9 Bilevel Optimization

A bilevel (Stackelberg) program optimises a *leader's* objective subject to a *follower* that itself solves an optimisation problem parameterised by the leader's decision.[60] When the lower level is convex in the follower variables (an LP or convex QP), it can be replaced by its Karush–Kuhn–Tucker conditions, collapsing the two-level problem into a single-level mathematical program with equilibrium constraints (Section 4.7). The `discopt.bilevel` module exposes a `BilevelProblem` builder that takes the leader model together with the follower's variables, objective, and constraints, and rewrites it in place under `formulate()`:

- **KKT reformulation** (`method`"kkt"=). The follower is replaced by its stationarity $\nabla_y L = 0$ (emitted symbolically over the expression DAG so the resulting constraints stay on the certifiable global path), primal and dual feasibility, and per-row complementarity $0 \le \mu_i \perp -g_i \ge 0$ handed to `discopt.mpec` under a selectable `mpec_method` (`gdp` or `sos1`).
- **Strong-duality reformulation** (`method`"strong_duality"=). The same stationarity and feasibility conditions are combined with a single aggregate strong-duality equality in place of per-row complementarity, avoiding disjunctive branching.

The builder handles the *optimistic* bilevel formulation and refuses loudly — rather than silently returning a wrong answer — on cases outside its soundness envelope: integer follower variables, a lower level nonconvex in y, or pessimistic semantics. Because both reformulations emit an ordinary `Model`, a bilevel program inherits the same spatial branch and bound, relaxations, and NLP backends as the rest of discopt.

4.10 DAG Consumers: Relaxation, Derivatives, and Differentiability

The expression DAG has three consumers, and it is worth being explicit about which one runs when, because the answer changed over the project's life and the earlier framing of this paper described a path that is no longer the default.

The uniform factorable pass (default). A single pass over the DAG lowers every node to the auxiliary-variable form of Section 7 and emits the relaxation as `numpy` arrays handed to the Rust simplex. There is no tracing, no compilation step, and no framework dependency; the same pass serves LP, QP, and the nonlinear envelopes, which is what lets the classifier of Section 4 route a linear model without paying for machinery it will not use.

The AD tape (default). Derivatives for node and incumbent NLP solves come from a tape built from the same DAG and evaluated inside POUNCE. This replaced a JAX evaluator that had been the single trigger importing JAX on every nonlinear solve; Section 17 reports the agreement measurements and the panel that licensed the switch.

The JAX compiler (optional). A separate compiler walks the tree and produces pure JAX functions compatible with `jax.jit`, `jax.grad`, and `jax.vmap`, mapping each node type (`Variable`, `Constant`, `BinaryOp`, `FunctionCall`) to a corresponding JAX operation. It supplies forward- and reverse-mode AD for the differentiable-solve layer (Section 10), parametric sensitivity, and the design-of-experiments Jacobians, and it is imported only when one of those is requested. Its properties — lazy evaluation until compilation, AD for gradients/Hessians/Jacobians, `jax.vmap` over batches of variable values or bound vectors, and XLA JIT for repeated evaluation — are what those subsystems need and what an ordinary solve does not.

4.11 Rust Backend

The Rust backend (`discopt-core` and `discopt-python` crates) provides three components:

Expression IR. A typed expression intermediate representation in Rust that mirrors the Python DAG but enables structure detection on the Rust side. PyO3 bindings[82] provide zero-copy data transfer between Python `numpy` arrays and Rust `Vec<f64>` via `into_pyarray()`.

Branch-and-bound tree. The `PyTreeManager` manages the B&B node pool in Rust with configurable selection strategies (best-first, depth-first). Node bounds, solutions, and parent solution vectors are stored in Rust memory. The Python layer calls `export_batch(k)` to retrieve k open nodes, dispatches the relaxations, and returns results via `import_results()`. This separation keeps tree manipulation in Rust with zero GIL contention while Python retains orchestration; the relaxation solves themselves land back in Rust, in the simplex for linear relaxations and in POUNCE for node NLPs.

Presolve. Feasibility-based bound tightening (FBBT) and probing propagate domain reductions through the expression graph. A `.nl` file parser enables importing problems from the MINLPLib benchmark library.[83]

5 NLP Solver Backends

discopt dispatches continuous NLP solves through a single seam that resolves to one of two interchangeable backends — POUNCE (the default) or Ipopt via cyipopt — so no call site needs to know which is in use. Both implement the same interface: given an objective function $f(x)$, constraints $g(x)$, variable bounds $x_l \leq x \leq x_u$, and constraint bounds $g_l \leq g(x) \leq g_u$, find x^* that minimizes f subject to all constraints.

A third backend, a pure-JAX interior point method, was the project's original NLP solver and has been retired. We describe it below because the *reason* it was retired is a result, not a footnote: it was written to make node relaxations JIT-compilable and `vmap`-able, it worked, it agreed with Ipopt to machine precision — and it was still the slowest of the three, at which point a Rust implementation of the same algorithm made the framework dependency unnecessary. Reporting only the surviving design would hide the measurement that decided it.

5.1 Pure-JAX Interior Point Method (retired)

The pure-JAX IPM implemented a primal-dual barrier method that was fully JIT-compilable and vmappable. It solved the augmented KKT system:

$$\begin{bmatrix} H + \Sigma + \delta_w I & J^T \\ J & -D - \delta_c I \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} = \begin{bmatrix} r_x \\ r_y \end{bmatrix} \tag{1}$$

where $H = \nabla^2_{xx}\mathcal{L}$ is the Hessian of the Lagrangian, $\Sigma = \text{diag}(z_l/s_l + z_u/s_u)$ is the barrier Hessian diagonal, J is the constraint Jacobian, D is the condensed slack diagonal for inequality constraints, and δ_w, δ_c are inertia correction regularizations.

5.1.1 Mehrotra Predictor-Corrector

Following Mehrotra,[18] each iteration consists of two linear system solves:

1. **Affine predictor** ($\mu = 0$): solve the KKT system with zero centering to obtain $(\Delta x^{\text{aff}}, \Delta y^{\text{aff}})$. Compute the affine complementarity μ^{aff}.
2. **Centering parameter**: $\sigma = (\mu^{\text{aff}}/\mu^{\text{curr}})^3$, clipped to $[0, 1]$.
3. **Corrector**: re-solve with centering $\sigma\mu$ and cross-product correction terms $\Delta s_i \Delta z_i / s_i$.

The corrector step provides second-order information about the central path, typically reducing the iteration count by 30–50\% compared to a single Newton step.

5.1.2 Inertia Correction

The KKT system must have correct inertia (n positive, m negative eigenvalues) for the Newton direction to be a descent direction. We detect incorrect inertia via Cholesky factorization: if the upper-left block $W = H + \Sigma + \delta_w I$ produces NaN in its Cholesky factor, the inertia is incorrect. The regularization δ_w is increased by a factor of 8 (up to $\delta_w^{\max} = 10^{40}$) until correct inertia is achieved. This Cholesky-based detection is approximately $3\times$ faster than eigenvalue decomposition.

5.1.3 Filter Line Search

Step acceptance follows Ipopt's filter line search.[16] The algorithm maintains an ℓ_1 merit function $\phi(x) = f(x) + \nu\|c(x)\|_1$ with adaptive penalty parameter ν and performs backtracking with Armijo-type sufficient decrease. A stall detection mechanism triggers projected steepest-descent recovery after 3 consecutive line search failures, handling the Maratos effect near active bound constraints.

5.1.4 Carry-Based `while_loop`

A critical implementation detail for JAX performance, recorded here because it is the kind of thing that is expensive to rediscover: all problem data (bounds, constraint bounds, masks) is placed in the `while_loop` carry state as a `NamedTuple`, rather than captured in the closure. Closure-captured data forces JIT retracing for each new problem instance, while carry-based data allows JIT cache reuse. Problem dimensions are derived from array shapes (`c.shape[0]`, `b.shape[0]`) rather than stored as integers, since integers in the carry become JAX tracers.

5.1.5 Batch Solving, and What Replaced It

Batched node evaluation was the reason the IPM was written in JAX at all. A `solve_nlp_batch` entry point applied `jax.vmap` over the initial-point and variable-bound dimensions:

```
def solve_nlp_batch(obj_fn, con_fn, x0_batch, xl_batch, xu_batch, g_l, g_u, opts):
    def _solve_single(x0, xl, xu):
        return ipm_solve(obj_fn, con_fn, x0, xl, xu, g_l, g_u, opts)
    return jax.vmap(_solve_single)(x0_batch, xl_batch, xu_batch)
```

solving all k node relaxations at once for roughly k-fold parallelism on hardware with sufficient FLOP throughput, with the objective and constraint functions shared across the batch and only the starting points and bounds varying per instance.

The interface survived the backend. `solve_nlp_batch` is now POUNCE's, dispatching the same $(node, start)$ pairs into Rust, and the contribution stands on the batching rather than on the framework that first expressed it. The batching *policy* is likewise unchanged: at the root node, multi-start solving uses 5 diverse starting points (midpoint, lower-quarter, upper-quarter, and 2 random) solved together; on non-root nodes the solver warm-starts from the parent's solution clipped to the child's tightened bounds.

5.1.6 Batching Whole Models

The batching above is *within* one solve: many node relaxations, one model. A second and independent axis is many independent models, and it is a decomposition rather than a parameter sweep. A CALPHAD phase diagram prices one phase per global solve and multiplies that by the number of temperature-composition conditions, giving tens to thousands of small, mutually independent global solves; the same shape arises in any multi-start, multi-scenario or per-cell decomposition. `dm.solve_batch(models, workers=n)` runs them in a process pool and returns one result per model in input order.

The interface is small, but two obstacles behind it are consequences of the Rust core and are worth recording, because every caller who wrote their own pool had to rediscover them. First, a `Model` that has been solved *cannot be pickled*: the solve leaves a live Rust representation and module references on the model and on its variables. Models are therefore shipped to workers as serialised text in the native round-trippable format of Section 16 and rebuilt on arrival. Second, a `SolveResult` carries a back-reference to its model so that `result.value(var)` works, which would drag the unpicklable model back through the pipe; the worker clears it and the parent re-attaches the caller's *own* model object on the way out, so a returned result behaves like one from a local solve.

What the batch deliberately does not promise is that a wall-clock `time_limit` means the same thing at every worker count: N solves sharing a machine each get less work done per second, so a budget-limited model can legitimately return a weaker bound under eight workers than under one. Models that terminate on *work* — the gap, or a node limit — return identical results either way, and those are the ones the equivalence test uses: on a 32-model panel of nonconvex bilinear programs, all 32 agree in status, objective and bound to 10^{-12} between sequential and parallel execution. The wall-clock gain on that panel is approximately twofold on four physical cores (three runs of the same panel gave 1.83×, 2.08× and 2.16×), which is reported rather than asserted, because the machine decides it.

Failure is isolated rather than fatal. A model whose solve raises comes back as an error result carrying its reason — the first place in discopt where a failed solve carries one — and the rest of the batch proceeds. The worker catches `BaseException` deliberately, because a panic from the native core does not derive from `Exception` and one model's panic must not take the batch with it. A result that refuses to pickle has its diagnostics dropped one at a time, worst-diagnostic-value first, each with a logged reason, on the principle that a result missing a diagnostic beats a lost solve; and a broken pool is re-raised with the explanation the pool's own message omits, namely that under `spawn` every worker re-imports the parent's `__main__` and an unguarded script re-runs its own batch inside each child.

5.2 POUNCE: A Memory-Safe IPM in Rust

POUNCE is a source-level translation of Ipopt's core interior point algorithm from C++ to Rust, developed as a separate companion project (https://github.com/jkitchin/pounce). It is **not** a wrapper around C++ Ipopt; it is a standalone Rust crate that produces native code with Rust's ownership and borrowing guarantees. discopt integrates POUNCE as one of its NLP backends through POUNCE's Python bindings (PyPI distribution `pounce-solver`, import name `pounce`).

5.2.1 Motivation

Three considerations drove the decision to translate rather than wrap:

1. **Numerical hardening.** Ipopt handles hundreds of edge cases accumulated over two decades: degenerate Jacobians, rank-deficient KKT systems, cycling detection, and tiny-step recovery. A from-scratch IPM would rediscover these failures one benchmark at a time. Translation preserves this institutional knowledge.

2. **Memory safety.** Rust's ownership model eliminates use-after-free bugs, data races, and buffer overflows at compile time—categories of bugs that have affected C++ numerical codes. The `cargo clippy` linter and optional `miri` interpreter catch additional issues.

3. **Deployment simplicity.** POUNCE compiles with `cargo build` with zero C or Fortran dependencies. The entire solver is a single Rust crate, eliminating the MUMPS/BLAS/LAPACK dependency chain that makes Ipopt installation non-trivial.

5.2.2 Architecture

POUNCE exposes a clean `NlpProblem` trait that users implement:

```rust
pub trait NlpProblem {
    fn num_variables(&self) -> usize;
    fn num_constraints(&self) -> usize;
    fn bounds(&self, x_l: &mut [f64], x_u: &mut [f64]);
    fn constraint_bounds(&self, g_l: &mut [f64], g_u: &mut [f64]);
    fn initial_point(&self, x0: &mut [f64]);
    fn objective(&self, x: &[f64]) -> f64;
    fn gradient(&self, x: &[f64], grad: &mut [f64]);
    fn constraints(&self, x: &[f64], g: &mut [f64]);
    fn jacobian_structure(&self) -> (Vec<usize>, Vec<usize>);
    fn jacobian_values(&self, x: &[f64], vals: &mut [f64]);
    fn hessian_structure(&self) -> (Vec<usize>, Vec<usize>);
    fn hessian_values(&self, x: &[f64], obj_factor: f64,
                      lambda: &[f64], vals: &mut [f64]);
}
```

The solver implements filter line search with Ipopt-style acceptance criteria, inertia correction via the `LinearSolver` trait (pluggable backends), feasibility restoration, warm-starting from previous solutions, and configurable convergence criteria.

5.2.3 PyO3 Integration

POUNCE ships PyO3 bindings[82] that discopt drives directly. A `PyNlpProblem` struct implements the `NlpProblem` trait by calling back into Python evaluator methods (`evaluate_objective`, `evaluate_gradient`, `evaluate_constraints`, `evaluate_jacobian`, `evaluate_lagrangian_hessian`). Bounds and sparsity structure are cached in Rust-owned memory. The main solve loop releases the Python GIL via `py.allow_threads()`, and callbacks reacquire it only for the duration of each evaluation:

```rust
let result = py.allow_threads(|| pounce::solve(&problem, &opts));
```

This means the heavy linear algebra in the IPM iteration runs without GIL contention. The callback proxy is itself now bypassed on the default path: for models that originate from (or can be emitted to) an `.nl`, POUNCE parses the problem into its own AD-backed representation once and differentiates natively, so the per-iteration Rust → Python boundary crossing disappears from the node loop entirely.

5.2.4 Licensing

Because POUNCE is a source-level translation of Ipopt (EPL-2.0), it is distributed under the Eclipse Public License 2.0 in its own repository (https://github.com/jkitchin/pounce). POUNCE replaced the earlier in-repo `ripopt` crate, which has been extracted into this standalone companion project. EPL-2.0 is not viral: code that *depends on* POUNCE does not need to adopt EPL-2.0. discopt depends on POUNCE only through its Python bindings; the retired pure-JAX IPM of Section 5.1 was also a re-implementation of the same algorithm, and we have adopted the EPL-2.0 license for the entire discopt repository to avoid confusion and ensure that all components are covered by the same license.

5.3 Ipopt via cyipopt

The second backend uses the industry-standard Ipopt solver[16] via cyipopt Python bindings. This serves as the correctness reference: any discrepancy between POUNCE and Ipopt on a well-conditioned problem indicates a bug in the former. Ipopt's advantage is its mature sparse linear algebra (MUMPS, MA27/MA57) which scales better on large, sparse problems ($n > 1000$). Its disadvantage is the Fortran/C++ dependency chain and inability to batch or differentiate through the solve.

5.4 Warm Starts and the KKT Residual Report

Two features that look separate ship as one, because one is the input the other produces. `Model.solve(warm_start=previous_result)` takes a whole `SolveResult` rather than a point, and splits it: the primal half becomes the ordinary starting point, so *every* route benefits from it, while the dual half — constraint multipliers, lower and upper bound duals, and the terminal barrier parameter — rides alongside and is converted into the interior-point solver's own warm-start object. Re-using duals is what distinguishes this from handing over a point. On a restricted-equilibrium NLP re-solved after a one-percent change in conditions, the same optimum to 10^{-7} took 7 interior-point iterations cold, 5 from the point alone, and 2 from the full primal–dual state.

The barrier parameter in that state comes from `SolveResult.kkt`, the second half of the pair: a report of the terminating KKT residuals — primal infeasibility, dual infeasibility, complementarity, the aggregate KKT error, and the terminal μ. Each residual is reported *twice*, once on the solver's internally scaled problem, which is what its own convergence test runs on and what its `kkt_error` means, and once in the model's own units, which is what a claim stated in problem units — a tangent-plane bound on a thermodynamic model, say — has to be built from. Both are reported rather than one silently standing in for the other. The field is `None` on every branch-and-bound route, because there is no single NLP there

whose residuals these would be, and a key the backend did not report is *omitted* rather than zero-filled, so a consumer sees a missing key instead of a number the solver never produced. Nothing in the solver reads the field back: it is diagnostic by construction, so a backend that omits a key cannot change a verdict.

The design rule shared by both halves is that a warm start which silently does nothing is the failure being removed, so every way of not using one is loud. Passing `warm_start` together with an explicit initial solution raises rather than picking a winner; passing a non-result raises; a streaming solve *refuses* a warm start outright, because the streaming driver accepts neither a start point nor duals and would take it inertly. Inside the conversion, a piece of dual state that does not fit this model's row layout is dropped and the solve proceeds from the point alone — but the mismatch is reported at warning level, never swallowed.

6 Pure-Rust LP/MILP Simplex Engine

The linear subproblems that dominate the global solver — the McCormick/AMP LP relaxation solved at every spatial-B&B node, and the MILP master of the AMP loop (Section 8) — are handled by a self-hosted, pure-Rust revised simplex rather than an external library. (A model classified as a *pure* LP or MILP at entry is a separate case and is routed elsewhere; see below.) A primal simplex handles cold and root solves; a bounded-variable *dual* simplex re-optimizes each B&B child from its parent's optimal basis, the warm start that makes node throughput competitive. The basis factorization is a pure-Rust sparse LU (the companion `feral` crate) with threshold partial pivoting and Forrest–Tomlin product-form updates between periodic refactorizations, so the engine carries no C or Fortran dependency and preserves the clean-wheel build.

Numerical robustness is handled *in-engine* rather than by swapping to a more robust external solver when a solve looks difficult. A numeric-focus mode adds iterative refinement of the basic solution and exposes element-growth and Hager–Higham condition estimates that gate a refined recompute of a drifted vertex before it is certified. Degeneracy — pervasive on the wide, lifted relaxations — is controlled by a Harris two-pass ratio test with an EXPAND expanding-tolerance minimum-step rule,[84] which reduces degenerate pivots by roughly 15× on the ill-conditioned lifted corpus, with a smallest-index (Bland) fallback that guarantees termination on a genuine cycle. Every `Optimal` is gated by an exact primal-feasibility audit and a Neumaier–Shcherbina safe dual bound, so a numerically drifted solve is downgraded to `Numerical` rather than trusted as a lower bound. This single hardened engine replaces an earlier fallback chain (external LP library → interior-point method → pure-JAX LP-IPM): every per-node LP relaxation in the spatial branch-and-bound now runs on the Rust simplex, and POUNCE is retained for linear work only where an interior analytic-center point is required, such as the KKT sensitivity of the differentiable-LP layer (Section 10).

A *pure* LP or MILP model — one classified at entry as having no nonlinear structure at all — is a separate case, and since v0.9.0 it does not use this engine by default. Such models are routed to HiGHS,[85] which is a core dependency rather than an optional one; `DISCOPT_LP_MILP_BACKEND=rust` is the opt-out that restores the Rust route, and an explicit `nlp_solver`"simplex"= still names it. The routing is deliberately confined to the class where a mature external code is strongest and where discopt's own relaxation machinery contributes

nothing, and it does not touch the per-node path that the global solver depends on.

Routing to an external solver would ordinarily widen the false-certificate surface, so the contract is *verify a certificate, do not trust a solve*: HiGHS's status labels are never trusted for a certified field. The returned point passes a readback guard against sentinel magnitudes, is verified feasible against the standard-form data with discopt's own residual kernel, and has its objective recomputed as $c^\top x$ rather than read from the solver; the dual bound is recomputed as a Neumaier–Shcherbina safe bound in Rust, and if that bound does not close, the result is reported as an honest feasible-but-uncertified incumbent rather than a false optimal. A verification failure withholds the point and raises — there is no silent fallback to another engine.

Two quantities are still taken on trust, and naming them is part of the contract. The MILP tree bound is HiGHS's own floating-point `mip_dual_bound`, carried through under an explicit `bound_provenance` label; this does not lower the provenance class of a pure-MILP certificate, because discopt's own MILP driver computes its optimality bound in floating point as well — the safe-bound routines are called from the spatial and convex B&B kernels, never from the MILP driver. And a bare infeasibility label is accepted as infeasible unless the root cross-check upgrades it to a verified Farkas certificate. Both are guarded from the outside rather than from within: after the tree solve, the root LP is re-solved through the same verified LP path, and a tree bound sitting below that safe root bound is an `error`, not a result. Where the guard cannot return a verdict — it was skipped for want of budget, came back inconclusive, or the root LP's own primal–dual pair fails a box-dual-feasibility check in unscaled arithmetic — the certificate is stripped rather than assumed: an `optimal` becomes an honest uncertified `feasible` keeping its incumbent, and an infeasibility claim, having no incumbent to fall back on, becomes an `error`. The design rule throughout is that the route *downgrades* rather than fails open.

7 Convex Relaxation Framework

The quality of convex relaxations at B&B nodes directly determines solver efficiency. discopt builds them with a compositional factorable pass that walks the expression DAG and emits `numpy` envelope data and LP rows.

7.1 McCormick Relaxations

The factorable pass implements 28 univariate and bivariate McCormick relaxation functions[19,20] covering all operations supported by the modeling API: bilinear products, addition, subtraction, negation, division, power, `exp`, square, `abs`, `sqrt`, `log`, `log2`, `log10`, `sin`, `cos`, `tan`, `atan`, `asin`, `acos`, `sinh`, `cosh`, `tanh`, `sigmoid`, `softplus`, entropy, `sign`, `min`, and `max`.

Each function maps interval bounds $[x_l, x_u]$ and convex/concave enclosures (cv, cc) of the argument to tighter enclosures of the result. The envelope algebra has one implementation: the uniform factorable pass emits `numpy` arrays and constraint rows straight into the Rust simplex. An earlier `jax.numpy` copy of the same rules served an alternative bounding route that solved an NLP over the compositional relaxation; on the in-repo corpus that route's bound changed the status, bound, or node count in none of 36 instance/variant pairs while

costing up to 27× the wall time, and it was retired together with its JAX envelopes (Section 17.4).

A uniform factorable relaxation engine. The per-operator relaxation table above is now driven by a single *uniform* factorable relaxation pass rather than a scattered per-operator dispatch. The earlier design attached bespoke code to each operator and to a "federation" of specialised claimers; that machinery was consolidated into one compositional walk that lifts every intermediate factor of the expression DAG, attaches its rigorous (cv, cc) envelope from the shared rule table, and assembles a single McCormick LP over the lifted variables. The consolidation deleted a large body of production-dead specialisation code while leaving the bound bit-for-bit identical on the certifying panel (a bound-neutral refactor in the sense of Section 15), and it is the substrate on which the composite atoms and cut separators below compose. The resulting LP relaxation is solved on the in-house Rust revised simplex of Section 6, not an external LP library, so an entire node — lift, relax, and solve — stays inside the hardened engine. A companion *adaptive* node-NLP policy (graduated default-on) tunes how aggressively each node's relaxation is re-solved.

Composite convex atoms. Beyond the scalar envelopes, the compiler recognises a set of multivariate *composite atoms* with known exact convex representations — log-sum-exp and softplus, the univariate entropy $x \log x$ and its relative-entropy generalisation, the Euclidean norm, and the convex $x\,e^x$ envelope on its convex domain. Each atom emits an exact convex outer approximation rather than the looser product-by-product McCormick fold, tightening the relaxation of the log-sum-exp, entropy, and norm structures that arise in geometric-programming and information-theoretic models.

7.2 User-Registered Composite Atoms

The composite atoms above are hand-written into the engine. A domain plugin, however, knows structure that generic factorable relaxation discards, and it cannot add to that list without modifying the solver. The motivating case is the Redlich–Kister binary excess Gibbs energy[86] that a CALPHAD phase-stability model[87] prices phases with,

$$g(x) = x(1-x)\Big(L_0 + L_1(2x-1)\Big) + RT\Big[x\ln x + (1-x)\ln(1-x)\Big], \qquad x \in [0,1]. \tag{2}$$

Written in primitives this is relaxed *term by term*: each product and each entropy term is enveloped over its own box and the pieces are summed, which loses every cancellation between them. `dm.register_function(name, lower)` lets a plugin name the composite so that the relaxer envelopes it as one atom.

The mechanism is a tag, not a node type. Calling a registered operator returns the ordinary expression its `lower` callable builds, carrying a Python attribute that names the atom it is the lowering of. Exactly one consumer reads that attribute: the canonicalisation pass that feeds the relaxer emits a single named call node instead of descending into the body. To the evaluator, the tape, the `.nl` exporter, presolve and the Rust core the model contains an ordinary expression and nothing changes — a registration adds no opcode, so nothing needs new support. That this holds is a test rather than a claim: the same test asserts that the Rust doorway accepts the model, that the tape agrees with the primitive

spelling to 10^{-12} at four points, that a registered *polynomial* atom exports to `.nl`, and that an atom whose lowering contains $x \log x$ still *refuses* to export, because `.nl` has no entropy opcode and registration must not become a way to bypass that refusal.

Nothing is taken on trust. `register_function` accepts no user-supplied envelope, so a registered operator cannot carry an unsound one — there is none to be unsound. The envelope is instead *derived* from the lowering. Value and first derivative are evaluated on a one-variable probe model through the same POUNCE tape the solver uses, so the envelope is built from the function the solver evaluates rather than from a second implementation of it. Curvature is decided per node box by evaluating the symbolic second derivative under interval arithmetic: an enclosure of f'' contains every value it takes on the box, so a sign verdict is a proof on that box rather than a sample, and an enclosure straddling zero abstains. On a verdict the atom enters the same secant/tangent path the built-in univariate atoms use; on an abstention the node keeps only its interval floor. Every degradation route — an unregistered name, a box outside the lowering's domain, an underivable registration, a node type the walk does not recognise — lands on that floor, which is the looser, always-valid relaxation. The domain guard is itself derived rather than hand-written: a box outside the natural domain produces a non-finite f'' enclosure, so the abstention arrives by the same evidence that produces every other one.

Two soundness defects were found and fixed in this mechanism, and both are worth stating because they delimit what "derived" buys. First, the symbolic differentiator takes *subgradients* at kinks, so f'' is blind to a kink's Dirac mass and $-|t|$ read as convex; the envelope then excluded the graph, the root relaxation of a feasible model was LP-infeasible, and separable concave minimisations certified above their optimum. The fix enumerates the lowering's kink sites and abstains on any box whose kink argument does not stay strictly on one side, and the walk *fails closed*: a node type it does not know raises, and the verdict abstains rather than assuming smoothness. Second, because the model keeps the body built by the registration that tagged it while the relaxer would derive its envelope from whatever the registry holds under that name now, a `replace`-ing registration could envelope an older model's body with a newer definition's envelope — measured cutting a true point by ≈ 100. The tag therefore carries the registration *object*, and a tag whose registration has been dropped or replaced degrades to term-by-term relaxation of the body the model actually carries.

An honest measurement. This mechanism was first reported here at 56–216× node-count reductions. That measurement is retracted: its primitive arm reached the relaxation engine's bare interval floor on every $x \log x$ term, because `entropy` — along with `softplus`, `sigmoid` and `tan` — was a first-class discopt intrinsic with no entry in the univariate envelope table. Almost the whole ratio was that missing envelope rather than the registration. Supplying the four missing entries is a separate contribution, and a larger one: it moves the root gap on a CALPHAD compound-energy formalism from 28.84\% to 5.45\%, on a coupled-sigmoid structure from 14.71\% to 0.00\%, on coupled `tan` from 84.75\% to 22.37\%, and on coupled `softplus` from 5.75\% to 2.41\%. Against a *repaired* baseline, naming the composite still earns its keep — a strict node-count reduction in every row of the family at the same optimum, 23–51 nodes falling to 15–47 — but by 1.09× to 1.53×, not by two orders of magnitude. The mechanism behind the modest figure is instructive: the Redlich–Kister binary is convex near the ends of $[0, 1]$ and concave in the middle, so on the root box the

derived verdict abstains exactly as it does for sin and cos, and the gain arrives only once branching has produced single-curvature boxes.

Why univariate only. Extending registration to multi-argument lowerings, with the curvature verdict becoming a Hessian enclosure, was proposed and falsified on 296 probes. The premise held: the canonicaliser flattens nested sums, so a multivariate composite written in primitives never exists as a node and registration is genuinely the only way to recover the grouping. The value did not. On the CALPHAD class that motivated it the convex lift never certified at any box width — 0/12 sub-boxes at each of six widths from 1.0 down to 0.005, against a convex control that fired 12/12 at every width — because a compound-energy cross term is bilinear and a bilinear Hessian $\left[\begin{smallmatrix} 0 & c \\ c & 0 \end{smallmatrix}\right]$ has eigenvalues $\pm|c|$ on *every* box, so an interval-Gershgorin bound can never be sign-definite. Where a composite did certify, its terms already carried tight envelopes and recovering the grouping changed the bound by -0.00%. And on $\log(e^x + e^y + e^z)$, the worst case for term-wise relaxation, grouping was worth 0.76 of a 107% root gap while the purpose-built log-sum-exp envelope closed it to 0.00% — a hand-written treatment, not something derivable from a lowering, and so precisely what this design excludes.

Registered atoms survive serialisation. Because the tag is a Python attribute, it does not cross a process or a file unless the document records it. It did not, and the path that lost it was exactly the parallel batch interface of Section 5.1.6, which ships every model to its worker as serialised text. A registered composite was therefore relaxed term by term in every worker, at no error and with nothing to see but a slower solve. The document now records which nodes are registered composites, but restoring the tag on read is not a matter of trusting a name: a name cannot carry across a process the object identity that protects against the replaced-registration false bound above. The reader instead re-derives it, re-lowering the *current* registration on the decoded argument and tagging only if that reproduces the decoded body exactly; every way of failing that check leaves the node untagged and relaxed term by term. Measured on a four-model batch of the Redlich–Kister family, the workers went from 23 nodes each — the term-by-term count — to 15, which is the count the same models take in-process.

7.3 α BB Underestimators

For general twice-differentiable functions where McCormick relaxations are loose, the α BB method[21,22] adds a separable quadratic perturbation:

$$L_\alpha(x) = f(x) - \sum_{i=1}^{n} \alpha_i (x_i - x_i^l)(x_i^u - x_i) \tag{3}$$

The parameters α_i are chosen such that L_α is convex: $\alpha_i \geq \max(0, -\lambda_{\min}(\nabla^2 f)/2)$ where $\lambda_{\min}$ is the minimum eigenvalue of the Hessian. The distinction that matters here is *rigorous versus sampled*, and the default is rigorous.

The per-node α used by the solver is computed from a sound interval enclosure of the Hessian with a per-row interval Gershgorin bound: for row i,

$$\lambda_{\min} \;\geq\; \underline{H_{ii}} - \sum_{j \neq i} \max\Big(|\underline{H_{ij}}|,\, |\overline{H_{ij}}|\Big)\,, \tag{4}$$

giving $\alpha_i = \max(0, -\frac{1}{2}\,\mathrm{gersh}_i)$, valid over the *whole* box rather than at sampled points. Variables that appear only linearly get $\alpha_i = 0$, so the perturbation is applied only where it is needed. Where the interval Hessian is unbounded the routine returns $+\infty$ and abstains — no α BB relaxation is emitted — rather than returning a number it cannot justify. This is `numpy` and interval arithmetic throughout, with no JAX import; the earlier arrangement, in which the per-node bound reached into a JAX module to get its α, was what dragged the whole framework onto an otherwise framework-free solve.

An earlier default took α from the Hessian *sampled* at interior points and inflated by 1.5×, with convexity checked only at the box centre. A negative-curvature band narrower than the sample spacing then left L_α nonconvex, its "lower bound" exceeded the box minimum, and the node holding the optimum was fathomed. That estimator has been removed rather than demoted: no sampled α reaches a dual bound, and a randomized panel of sub-boxes of exactly that objective pins the bound below a dense-grid box minimum.

7.4 Piecewise McCormick Relaxations

Piecewise McCormick[23] partitions the variable domain into k subintervals and constructs tighter McCormick envelopes on each partition, at the cost of binary variables that select the active subinterval. In discopt this is the relaxation of the AMP solver (Section 8), which starts from a uniform partition and refines it adaptively by adding breakpoints near the current MILP solution, where the relaxation gap is largest. The spatial branch-and-bound engine does not partition; it tightens by branching, and a former `partitions` option that partitioned inside its compositional NLP relaxation is now refused with an error.

On representative univariate test functions, piecewise McCormick with $k = 8$ uniform partitions reduces the mean envelope gap $\overline{cc - cv}$ over the interval by 97–98% (`manuscript/piecewise_gap_reduction.ipynb`, 10,000 uniform samples per function; each sampled mean agrees with its closed-form value to within 3%). For x^2 the reduction is exactly $1 - 1/k^2$:

Function	Interval	Standard gap	Piecewise gap	Reduction
$\exp(x)$	$[-2, 3]$	6.110	0.127	97.9%
$\log(x)$	$[0.5, 8]$	0.568	0.017	97.0%
x^2	$[-3, 4]$	8.138	0.128	98.4%

Table 2: Mean envelope gap $\overline{cc - cv}$, standard versus piecewise McCormick with $k = 8$ uniform partitions.

7.5 Cutting Planes: RLT, SOC, and Moment/PSD Separators

Envelope-based relaxations bound each nonlinear term in isolation; cutting planes tighten the relaxation by exploiting the **joint** structure of several terms. discopt separates four cut families, each as a `numpy` separator that returns violated inequalities at the current relaxation point:

- **Reformulation-linearization (RLT).**[46] Level-1 RLT multiplies pairs of constraint and bound factors and linearises the resulting products against the lifted bilinear variables, tightening the root bound on quadratically-constrained problems; the same products are also separated per node as targeted RLT cuts. The recursive McCormick fold for n-ary products is augmented by exact multilinear convex-envelope cuts[88] up to a capped arity.

- **Second-order-cone (SOC)** cuts. For a rotated or standard cone $\|y\|_2 \le t$, gradient (outer-approximation) cuts at the relaxation point provide a polyhedral outer approximation that sharpens the bound without a conic solver. The SOC and moment/PSD separators together target the quadratically-constrained relaxations that arise in alternating-current optimal power flow,[89] which serves as a capstone stress test for the cut framework.

- **Moment / PSD eigenvalue** cuts.[47] For box-QP and QCQP substructure, the lifted moment matrix $M(x, X) = \left(\begin{smallmatrix} 1 & x^\top \\ x & X \end{smallmatrix}\right)$ must be positive semidefinite; the separator computes its most negative eigenvalue and adds the corresponding eigenvector cut $v^\top M v \ge 0$, the first-order semidefinite relaxation.

- **Cover and clique** cuts for 0–1 knapsack substructure.

Rather than expose this machinery as a tuning burden, discopt defaults to a structure-gated policy: `Model.solve(cuts"auto")`= inspects the detected problem class and separates RLT on quadratically-constrained problems that carry linear constraints, PSD cuts on constraint-free box-QPs, and neither when no structure is present, so that the two strong but redundant families are never run together. An explicit `rlt_cuts`, `psd_cuts`, or `rlt=True/False` flag overrides the policy for controlled experiments, and `cuts"manual"`= disables it entirely.

Two further root-bound strengtheners target constrained binary and quadratically-constrained problems specifically. For a constrained binary quadratic program, an exhaustive level-1 RLT root bound is computed and, on the coupling rows, taken to its *Lagrangian dual* so that the RLT multipliers are optimised rather than fixed; the resulting bound is passed through a Neumaier–Shcherbina safe-rounding step before it is trusted as a dual bound, and a bound-neutral set-partitioning exclusion presolve removes RLT products that a set-partitioning constraint already fixes. Separately, root *optimality-based bound tightening* (OBBT) iterates min/max probes of each variable over the McCormick LP relaxation to a fixpoint, tightening the box from which every downstream envelope is built; this is the QCQP-root complement to the FBBT of Section 4 and is exposed as part of the branch-and-reduce root pass. Both are held behind graduation flags until a differential panel certifies them cert-clean and net-positive.

An exact-linearisation route handles binary-multilinear structure without any envelope at all: a product of $\{0, 1\}$ variables (as in autocorrelation and Boolean-polynomial models) is replaced by its exact McCormick-tight linearisation, converting the node relaxation into an equivalent MILP that the Rust simplex/branch-and-bound solves to global optimality directly, with incumbent seeding to warm the search.

7.6 Structural Convexity Certification

Many problems written in unconstrained nonlinear form are in fact convex — a sum of squares, a log-sum-exp, an exponential cone, or an affine function with a convex regulariser — but the modelling layer cannot exploit that structure unless it is **proved**. discopt ships a sound convexity detector under `discopt._relax.convexity`, inspired by the disciplined convex programming (DCP) framework of Grant, Boyd and Ye[24] and the SUSPECT special-structure detector of Ceccon, Siirola and Misener.[25]

The detector classifies each (sub)expression by curvature (CONVEX, CONCAVE, AFFINE, or UNKNOWN) using two complementary, **sound** methods. The syntactic walker (`rules.py`) implements DCP composition rules: an affine combination of convex expressions with non-negative weights is convex; a non-decreasing convex function of a convex expression is convex; and so on. When the syntactic walker abstains, the box-local certificate (`certificate.py`) closes the gap by computing a sound interval enclosure of the Hessian over the variable box via interval automatic differentiation (`interval_ad.py`) and a sound lower bound on the minimum eigenvalue across that enclosure (`eigenvalue.py`, with both a Gershgorin disc bound and a tighter Schur-complement bound for arrow-structured Hessians); a non-negative lower eigenvalue bound proves convexity by the second-order sufficient condition.[22] Symmetrically, a non-positive upper bound proves concavity. Any other outcome returns a conservative abstention — the certificate never claims nonconvexity, and never loosens a verdict from the syntactic walker.

Three downstream consumers benefit from a CONVEX verdict:

- **Convex fast path.** A model whose objective and all constraints certify CONVEX (or AFFINE) is routed to a single NLP solve with =SolveResult.convex_fast_path = True=, skipping spatial branch and bound entirely. For pure convex QPs this dispatches to POUNCE with a quadratic cost (the QP path is POUNCE-only), with the same status semantics as the general MINLP solve. Soundness of the certificate guarantees that the single NLP optimum is the global optimum.

- **Tighter αBB.** The same interval-Hessian eigenvalue bound drives the rigorous α of Section 7.3, so on terms that certify CONVEX it gives $\alpha = 0$ and the underestimator is the function itself.

- **Multi-objective scalarization** (Section #sec-mo) inherits the same convex fast path on each subproblem when the scalarised model certifies convex, e.g. weighted-sum on a sum of certified-convex objectives with non-negative weights.

The detector is conservative by design: it underclaims rather than overclaims. Every CONVEX or CONCAVE verdict is a mathematical proof, never a heuristic; sampling-based methods that could produce false positives are not used.

7.7 Geometric Programs

A geometric program (GP)[51,52] minimises a posynomial subject to posynomial-$\leq$-monomial and monomial-=-monomial constraints over strictly positive variables. A GP is generally

nonconvex in the original variables, so a curvature certificate in x-space (Section 7.6) abstains; yet under the change of variables $y = \log x$ every monomial becomes the exponential of an affine form and every posynomial a convex sum of exponentials, rendering the whole program convex. discopt treats this as a distinct, sound verdict — **log-convexity** — computed by `discopt.gp.is_log_convex` without disturbing the x-space convexity detector.

When `classify_gp` recognises the standard GP form, `solve_model` routes the problem automatically through `as_geometric_program`, which builds the log-space convex NLP, solves it once with the convex fast path (a single global solve, no spatial branch and bound), and maps the optimiser back to x-space; soundness of the log-convex reformulation guarantees the recovered point is the global optimum. The routing is automatic for recognised GPs but can be forced with `solver`"gp"= or bypassed with `solver`"bb"=, and it is suppressed when streaming callbacks require the branch-and-bound path. Recognising log-convex structure converts an otherwise nonconvex global-optimisation problem into a single convex solve, the same payoff the convexity certifier delivers for x-space-convex models.

Generalised (G-)convexity. The log-space transform is one instance of a broader *generalised-convexity* recognition layer. A per-expression log-curvature lattice is propagated over the DAG, and a G-convexity detector applies an augmented-Hessian test to certify convexity under a monotone reparameterisation even when neither the x-space certifier nor the strict GP recogniser fires. Where a sub-expression is G-convex (or, dually, log-concave) but the model as a whole is not, the layer emits rigorous *transformation-supporting* cuts — a signomial secant overestimator, a log-concave overestimator gated by a soundness proposition, and product/ratio envelopes in the transformed space — that tighten the node relaxation without disturbing the x-space bound. Each cut family is gated behind the differential-bound + feasible-point regime of Section 15 and injected only after passing a cert-clean panel.

GP-structured MINLP. When a geometric-program core carries genuine integer or binary variables, the pure log-space convex solve no longer applies, because the change of variables $y = \log x$ is defined only on the continuous positive block. discopt handles this *GP-structured MINLP* class by branching on the integers while solving each node's continuous relaxation in y-space: the integer variables drive a spatial/combinatorial branch-and-bound tree, and every node relaxation is the log-space convex program over the continuous GP block, certified by the same log-convex soundness argument. This routes an integer-augmented posynomial model to a sound global certificate rather than to a local NLP, and is gated behind a graduation-panel flag pending its net-positive graduation.

8 Adaptive Multivariate Partitioning (AMP)

In addition to the spatial branch and bound described in earlier sections, discopt provides AMP — a complementary global MINLP solver based on the Adaptive Multivariate Partitioning algorithm of Nagarajan and co-workers,[28,29] also known as the algorithm behind the Alpine solver. AMP is selected with `Model.solve(solver`"amp")= and is preferred when the bilinear / multilinear structure of the problem benefits from refined piecewise relaxations rather than from binary branching.

Where sBB enumerates a binary tree of variable-bound subregions, AMP iterates a single tightening loop on the **partitioning** of the continuous-variable domain. At iteration k the

algorithm performs three steps:

1. Build a piecewise-McCormick MILP relaxation of the original MINLP using the current partition $\mathcal{P}^{(k)}$ of each branched variable, solve it to optimality, and obtain a lower bound LB_k together with a candidate point $\hat{x}^{(k)}$.

2. Fix each branched variable's interval assignment from $\hat{x}^{(k)}$ and solve the remaining continuous NLP to local optimality, obtaining an upper bound UB_k if feasible.

3. If $(UB_k - LB_k)/|UB_k| \le \varepsilon$, certify global optimality and stop. Otherwise refine $\mathcal{P}^{(k)}$ by splitting the partition cell containing $\hat{x}^{(k)}$ along its longest edge, increment k, and repeat.

Soundness is maintained at every iteration by the McCormick guarantee on the MILP relaxation:

$$LB_k \;\le\; \min_{x \in \mathcal{X}^{\text{MINLP}}} f(x) \;\le\; UB_k \quad \text{for all } k, \tag{5}$$

so termination at the first k with closed gap is a certified ε-optimal solution. The MILP-relaxation construction is shared with the convex-hull / Big-M formulation work used elsewhere in discopt and is implemented in `discopt._relax.milp_relaxation`. The MILP itself is solved by the self-hosted pure-Rust simplex engine (Section 6), and the NLP subproblem by either NLP backend (default `pounce`; cyipopt is selected automatically when available).

Compared with the default sBB, AMP trades branching combinatorics for relaxation tightness: the MILP at each iteration is larger (one binary per partition cell per branched variable) but the tree is shallow because partitions, not subregions, are refined. On problems where the optimum lives on a few active partition cells — pooling problems, water-network synthesis, generalised disjunctive programs with bilinear terms — AMP closes the gap in tens of iterations where sBB would explore thousands of nodes.

The pooling problem[90,91] — blending streams of differing quality through intermediate pools to meet output specifications — is the canonical bilinear test of relaxation strength. The `discopt.pooling` module builds the **pq-formulation** via `build_pq_formulation`, which introduces proportion variables $q_{i\ell}$ (the fraction of pool ℓ's inflow drawn from input i) together with reformulation-linearization equalities $\sum_i w_{i\ell j} = y_{\ell j}$ for the bilinear products $w_{i\ell j} = q_{i\ell}\, y_{\ell j}$. These redundant-but-tightening constraints make the pq relaxation provably at least as tight as the textbook p- and q-formulations; on the classic Haverly instances (`haverly_hpp1`) the root relaxation is already exact, so the global optimum is certified at the root node.

v0.4 hardening. The first AMP release covered the core piecewise-McCormick / disaggregated-convex-hull path on bilinear and trilinear terms. The v0.4 line extends term coverage and relaxation tightness in four directions:

1. **Fractional-power lifting.** Fractional powers x^p with $p \in \mathbb{Q} \setminus \mathbb{Z}$ are lifted to MILP auxiliary variables with piecewise-secant under- and over-estimators, removing the previous restriction to integer monomial degrees.

2. **Universal piecewise-secant coverage.** Concave univariate terms (log, $\sqrt{\cdot}$, fractional powers, concave exponential branches) now receive piecewise-secant outer approximations consistent with the rest of the relaxation, rather than fall back to a single tangent.

3. **β-driven piecewise McCormick.** For products of the form $x \cdot y$ where one factor itself contains a fractional-power expression, AMP refines the partition along the variable that drives the largest envelope gap rather than the longest edge, giving faster gap closure on signomial pooling instances.

4. **Optimisation-based bound tightening on the relaxation.** An opt-in OBBT pass on the active MILP relaxation tightens variable bounds before each MILP solve. When an incumbent is available, the cutoff is added as a constraint during OBBT, and live partition state `disc_state` is passed through so the tightening reflects the current $\mathcal{P}^{(k)}$ rather than the root box.

The convex-hull formulation can be selected via `convhull_formulation \in \{=disaggregated, sos2, facet\}=` with the optional logarithmic Gray-code embedding `convhull_ebd=True` (giving $\log_2 |\mathcal{P}|$ binaries per branched variable instead of $|\mathcal{P}|$). The algorithm also exposes `rel_gap`, `max_iter`, `time_limit`, `n_init_partitions`, `partition_method` (`adaptive`, default; `longest-edge`; `equal-bisection`), `presolve_bt`, `obbt_at_root`, `obbt_with_cutoff`, `alphabb_cutoff_obbt`, and an `iteration_callback` hook receiving `(iteration, lower_bound, upper_bound)` for live convergence inspection on `Model.solve(solver"amp", ...)=`.

9 Branch-and-Bound Search Enhancements

The lower bound at each node is only half of branch and bound; a strong incumbent prunes the tree, a good node ordering reaches it sooner, and learning from infeasible subtrees avoids repeating mistakes. discopt implements a set of search enhancements that, by construction, never weaken the global bound or the optimality certificate.

Primal heuristics. Beyond the feasibility pump,[63] `discopt._relax.primal_heuristics` provides three improvement heuristics that take the relaxation solution (and, where applicable, the incumbent) and return candidate integer-feasible points. **Diving** progressively fixes one fractional integer at a time and re-solves the node NLP, in either a fractional (nearest-integer) or objective-guided rounding mode. **RINS**[92] fixes every integer on which the incumbent and the relaxation agree and searches the small sub-MINLP over the disagreeing integers. **Local branching**[93] adds a k-flip Hamming-distance constraint around the incumbent and solves the resulting neighbourhood exactly. All three only propose incumbents; every bound mutation they make is temporary and restored, so they cannot affect the dual bound. They are opt-in and invoked at controlled points in the search rather than run unconditionally.

Node selection. In addition to best-first and depth-first, discopt offers a best-estimate strategy that orders open nodes by a pseudocost-style estimate of the integer optimum reachable below them, and an objective-gating priority rule that biases branching toward variables most likely to raise the dual bound. Because selection only reorders an exhaustive

enumeration, any strategy preserves correctness; the strategies differ only in how quickly the gap closes.

Conflict analysis. When a node is proved infeasible, `discopt.conflict` analyses the responsible bound changes to extract a minimal infeasible assignment of binary variables and emits a no-good cut that excludes it tree-wide.[54] Because the excluded assignment is genuinely infeasible, the cut removes no feasible point and the global bound is unaffected.

Bound tightening as a public API. The feasibility-based bound tightening used internally in presolve is also exposed directly as `discopt.tightening.fbbt_box(model)`, which returns tightened variable bounds (or a proof of infeasibility when constraint propagation yields an empty interval) via row-activity interval arithmetic with no LP solve, supporting both standalone model analysis and reuse inside custom search loops.

Infeasibility analysis. For models that are proved infeasible, `discopt.infeasibility.compute_iis` returns an irreducible infeasible subsystem (IIS): a subset of constraints and variable bounds that is itself infeasible but becomes feasible if any single member is removed. The implementation uses deletion filtering,[53] tentatively dropping one member at a time and retaining it only when the remainder is **provably** infeasible. This guard makes the procedure sound under an incomplete solver: the returned set is always genuinely infeasible, and it is certified irreducible exactly when every drop-probe was conclusive. Together with the convergence and limit diagnostics, the IIS turns a bare `INFEASIBLE` status into an actionable, human-readable explanation of which constraints conflict.

10 Differentiable Optimization

A distinguishing feature of discopt is the ability to differentiate through the optimization solve with respect to problem parameters. This enables decision-focused learning[12] where the model parameters (e.g., cost coefficients) are learned end-to-end from decision outcomes rather than prediction accuracy.

10.1 Level 1: Envelope Theorem

For a parametric NLP $\min_x f(x;p)$ subject to $g(x;p) \le 0$, the envelope theorem gives:

$$\frac{df^*}{dp} = \frac{\partial \mathcal{L}}{\partial p}\bigg|_{x^*,\lambda^*} = \frac{\partial f}{\partial p}\bigg|_{x^*} + \lambda^{*T}\frac{\partial g}{\partial p}\bigg|_{x^*} \tag{6}$$

This requires no additional linear system solve—the optimal dual variables λ^* from the NLP solver directly provide the sensitivity. discopt implements this as a `DiffSolveResult.gradient(param)` method that returns df^*/dp for any model parameter.

10.2 Level 3: Implicit Differentiation via KKT

For the full primal sensitivity matrix dx^*/dp, discopt differentiates the KKT conditions:

$$\begin{bmatrix} \nabla^2_{xx}\mathcal{L} & J_a^T \\ J_a & 0 \end{bmatrix} \begin{bmatrix} dx^*/dp \\ d\lambda^*/dp \end{bmatrix} = - \begin{bmatrix} \nabla^2_{xp}\mathcal{L} \\ \partial g_a/\partial p \end{bmatrix} \tag{7}$$

where J_a is the Jacobian of active constraints (including active variable bounds) and $\nabla^2_{xp}\mathcal{L}$ is the mixed Hessian of the Lagrangian. This is computed via `jax.hessian` and `jax.jacobian` applied to parametric DAG-compiled functions.

The `DiffSolveResultL3` class provides:

- `sensitivity_matrix()`: full dx^*/dp matrix
- `implicit_gradient(param)`: df^*/dp via the chain rule $\nabla_x f \cdot dx^*/dp + \partial f/\partial p$
- `approximate_resolve(new_params)`: first-order approximation $x^* + (dx^*/dp)\Delta p$
- `dual_sensitivity(param)`: $d\lambda^*/dp$ for constraint multiplier sensitivity
- `reduced_hessian()`: Hessian projected onto the null space of active constraints (for uncertainty quantification)
- `sensitivity(dp)`: fast re-query with cached KKT factorization (analogous to sIPOPT[70])

If the KKT system is ill-conditioned, discopt falls back to finite-perturbation smoothing, and then to L1 envelope theorem gradients.

10.3 Differentiable Mixed-Integer Programs via Fix-and-Differentiate

The envelope and implicit-KKT sensitivities above assume a continuous solution manifold, which the integer variables of a MILP, MIQP, or MINLP break. discopt extends differentiability to the mixed-integer case through a **fix-and-differentiate** restriction: the problem is first solved to global optimality, the integer variables are then pinned at their optimal values, and the continuous restriction — now an ordinary parametric NLP — is differentiated by the Level 1 or Level 3 machinery above. This is exact wherever it matters: the optimal integer assignment is piecewise-constant in the parameters, so on the (open, full-measure) set where that assignment is locally stable the objective is smooth through the continuous restriction and df^*/dp is the true gradient; at the measure-zero breakpoints where the assignment switches, the procedure returns the one-sided sensitivity of the incumbent assignment, which is the appropriate convention for the gradient-descent updates of decision-focused learning.

A single entry point `differentiable_solve(model, method"auto")`= detects the problem class (LP, QP, NLP, MILP, MIQP, MINLP) and dispatches automatically, returning a `UnifiedDiffResult` that carries the optimal objective and solution, the detected `problem_class`, a `gradient(param)` method, and the continuous-relaxation objective for mixed-integer problems. The continuous paths reduce to the envelope/implicit-KKT results of the preceding subsections, so a user differentiating a model need not know in advance whether it contains integers.

10.4 Embedding an Inner Optimization: `argmin` Blocks

The sensitivities above differentiate *a* solve. An `argmin` block instead makes the minimizer of an inner model available as an object of an outer one: $\varphi(p) = \arg\min_y \{f(y;p) : g(y;p) \le 0\}$, used inside an outer objective or constraint, or standing alone as a twice-differentiable layer.

discopt exposes two arms, and the choice between them is the choice this section is about — it is made explicitly rather than by a lowering that changes underneath the user.

`dm.argmin(inner, bind={param: outer_expr})` embeds the inner `Model` as an opaque node. Its forward pass is a POUNCE solve of the inner problem at the current parameter values, and the node's value is the minimizer POUNCE returns. Its derivatives come from the sIPOPT sensitivity system, but wired in through `jax.lax.custom_root` over the inner KKT residual rather than returned as arrays, so first *and* second order both work — the outer solver's Lagrangian Hessian is a forward-over-reverse pass through the node — and the right-hand side is exact, differentiated symbolically off the compiled DAG rather than finite-differenced. The block admits any follower discopt can solve, and pays for that generality by reporting a *local* answer: the outer solve is a local NLP with `status"feasible", =gap_certified=False` and no dual bound, and the model cannot be exported to `.nl`.

`dm.argmin_kkt(model, inner, bind...)=` takes the opposite trade: it lowers the follower *into* the outer model, so the follower's variables become real columns and its optimality conditions become real rows, and the whole problem goes through the ordinary stack — Rust tape, FBBT, spatial branch and bound, and a global certificate. Under `method"strong_duality"=` the emitted rows are pure algebra (stationarity, primal and dual feasibility, and the single bilinear equality $\sum_i \mu_i g_i = 0$), so the model writes a valid `.nl` file and the identical formulation can be handed to another solver; under the default `method"kkt"=` the complementarity rows are disjunctive or SOS1, which discopt certifies but for which `.nl` has no row, and an export is refused by name rather than silently approximated.

What the lowering costs is generality, and the cost is not negotiable: KKT conditions characterise a follower's optimum only when the follower is convex in its own variables. For a nonconvex follower the KKT set contains every stationary point, so the leader is free to select the follower's *maximizer* whenever that suits the leader's objective — and the solver will certify it, because that point genuinely is optimal for the model as written. On a projection follower $\min_y \|y-(p,1)\|^2$ s.t. $\|y\|^2=1$ with the leader asking for $y_0=-0.9$, a hand-written KKT block returns the far-side root $y^*=(-0.9,-0.436)$ and certifies it; it is not the projection of anything. `argmin_kkt` therefore runs the convexity certifier of Section 4.9 and refuses any follower it cannot prove convex in y — LP, convex-QP and certified convex-NLP followers pass, a nonlinear equality does not — and there is no flag to override that. An unprovable follower goes through `dm.argmin`, which solves rather than reformulates and reports its answer as local.

The opaque arm's own gates refuse rather than approximate, and because a traced solve cannot raise, the refusal mechanism is a NaN that the outer NLP reports as a failed evaluation: a failed inner solve, an inner KKT residual that does not check out at the returned multipliers, or a reduced Hessian $Z^\top W Z$ that is not positive semidefinite on the null space of the active set. That last gate is exactly the failure the hand-written route suffers. A degenerate active set — an active row whose multiplier is below tolerance — is warned rather than silenced, because dy^*/dp is only one-sided there and the outer gradient should be read as a subgradient. Statically, an inner model with no objective, a maximizing inner model (which would be an argmax with an inverted sign convention), integer inner variables (whose optimum KKT stationarity does not characterise), a non-scalar bound parameter, or an empty `bind` are each rejected with a message naming the alternative. Measured on a

follower where the two arms overlap, the lowered and opaque answers agree to 10^{-4}, with the lowered arm returning `optimal` and `gap_certified=True` where the opaque arm returns `feasible` with no bound; on a problem with a known answer, the opaque arm lands on the optimum a 20,001-point brute-force scan identifies, to 10^{-3} in the parameter and 10^{-6} in the objective.

10.5 Application: Decision-Focused Learning

The JAX-native differentiable solve is exposed via `jax.custom_jvp`, enabling it to be embedded in a larger JAX computation graph. A decision-focused learning pipeline:

1. A neural network $\hat{c} = \phi_\theta(x_{\text{features}})$ predicts cost parameters
2. The predicted costs are fed to `discopt.solve()` to obtain decisions $x^*(\hat{c})$
3. The true task loss $\ell(x^*(\hat{c}), c_{\text{true}})$ is computed
4. Gradients $\partial \ell / \partial \theta$ flow through the solver via L3 implicit differentiation
5. The network parameters θ are updated to minimize decision regret

On a resource allocation task with 3 activities, 5 features, and risk-averse objective, decision-focused learning with discopt achieves **36% regret reduction** compared to two-stage predict-then-optimize, despite having higher prediction MSE. This confirms the key insight: optimizing for decision quality beats optimizing for prediction accuracy.

On a mean-variance portfolio optimization task with 3 assets and nonlinear interactions, DFL similarly achieves lower regret and higher utility than the two-stage approach.

11 Dynamic Optimization

Dynamic optimization problems (optimal control, trajectory optimization, nonlinear model predictive control) are solved in discopt by transcribing ODE/DAE systems into algebraic NLP constraints using orthogonal collocation on finite elements.[72,73] The `discopt.dae` module provides a `DAEBuilder` high-level interface: users declare state, algebraic, and control variables, supply an ODE/DAE right-hand side as a Python callable that builds discopt expressions, and call `discretize()` to generate all collocation and continuity constraints automatically.

Given a dynamic system $\dot{x} = f(x, z, u, t; \theta)$ with algebraic variables z satisfying $0 = h(x, z, u, t; \theta)$, each finite element $[t_k, t_{k+1}]$ of length h_k is transcribed by placing K collocation points $\tau_j \in (0, 1]$, introducing stage variables $x_{k,j}$, $z_{k,j}$, and enforcing

$$\sum_{l=0}^{K} \dot{\ell}_l(\tau_j)\, x_{k,l} = h_k\, f(x_{k,j}, z_{k,j}, u_k, t_{k,j}; \theta), \quad j = 1, \ldots, K, \tag{8}$$

$$0 = h(x_{k,j}, z_{k,j}, u_k, t_{k,j}; \theta), \tag{9}$$

$$x_{k+1,0} = \sum_{l=0}^{K} \ell_l(1)\, x_{k,l}, \tag{10}$$

where ℓ_l are Lagrange basis polynomials on the collocation nodes. Radau IIA (1–5 points) and Gauss-Legendre schemes are supported, along with automatic state augmentation for second-order ODEs and index-1 DAEs with algebraic constraints enforced at each collocation point. Piecewise-constant controls u_k and integral objectives $\int L(x, u, t)\, dt$ via Gauss quadrature cover standard optimal-control formulations. A finite-difference backend (`FDBuilder`) provides backward, forward, and central Euler stencils as a simpler alternative for users who prefer explicit time-stepping.

This transcription approach requires no changes to the DAG compiler or solver infrastructure: the collocation constraints are standard algebraic expressions built from existing operators, so a transcribed model takes the ordinary relaxation and AD-tape path, and is equally available to the differentiable-solve machinery of the Differentiable Optimization section, without special-casing in either direction. For a 30-element, 3-point Radau discretization of $\dot{x} = -x$ with $x(0) = 1$, the solution matches `exp(-t)` to within 10^{-4} absolute error on $[0, 10]$. The vectorized collocation evaluator (introduced in a recent NMPC-oriented perf update) computes stage-residuals for all elements and collocation points in a single array-shaped kernel, eliminating Python-level loops over the horizon.

Because collocation reduces a dynamic problem to a structured sparse NLP, the same transcription machinery supports dynamic parameter estimation (Section 12) and dynamic experiment design (Section 13): the parameter vector θ enters f, h as an ordinary unknown, and the FIM through the dynamics is computed by the same parametric autodiff used for the sensitivities of Section 10.

12 Parameter Estimation

The `discopt.estimate` module estimates unknown model parameters from experimental data using weighted nonlinear least squares (equivalently, Gaussian-noise maximum likelihood).[94] Given responses y_i with measurement standard deviations σ_i, the estimation problem is

$$\hat{\theta} = \arg\min_{\theta \in \Theta} \sum_{i=1}^{N} \left(\frac{y_i^{\text{obs}} - y_i^{\text{model}}(\theta; d_i)}{\sigma_i} \right)^2, \tag{11}$$

where d_i are design conditions (sampling times, input concentrations, etc.) for observation i, Θ is a parameter box, and y_i^{model} is any discopt expression, including one produced by the DAE transcription of Section 11. Because the objective is an ordinary discopt expression, estimation reuses the full NLP stack: POUNCE or cyipopt, over the same AD tape.

Users declare an `Experiment` by subclassing a thin interface and providing `create_model(**kwargs)` returning an `ExperimentModel` that tags which variables are unknown parameters, which are design inputs, which expressions are responses, and the per-response measurement error. The same `Experiment` object is reused across estimation, FIM computation, and optimal experimental design, ensuring the three share a single source of truth.

```
from discopt.estimate import Experiment, ExperimentModel, estimate_parameters

class ArrheniusExperiment(Experiment):
    def create_model(self, **kwargs):
```

```
        m = dm.Model("arrhenius")
        A  = m.continuous("A",  lb=1e3,  ub=1e8)
        Ea = m.continuous("Ea", lb=1e4,  ub=1e5)
        T  = m.continuous("T",  lb=300,  ub=500)
        k  = A * dm.exp(-Ea / (8.314 * T))
        return ExperimentModel(
            model=m,
            unknown_parameters={"A": A, "Ea": Ea},
            design_inputs={"T": T},
            responses={"k": k},
            measurement_error={"k": 0.05},
        )

result = estimate_parameters(ArrheniusExperiment(), data, initial_guess={"A": 1e6, "Ea": 5e4})
print(result.summary())  # parameters, standard errors, 95% CIs, FIM det and cond
```

The `EstimationResult` reports point estimates, the asymptotic covariance $\Sigma_\theta \approx (J^\top \Sigma^{-1} J)^{-1}$ where $J = \partial y^{\text{model}}/\partial\theta$, standard errors, confidence intervals via Student's t with $N - p$ degrees of freedom, and the correlation matrix. The Jacobian J is computed by `jax.jacobian` on the parametric DAG-compiled response functions, giving exact sensitivities (no finite-difference noise) at negligible overhead relative to the estimation solve. For dynamic models, the collocation transcription of Section 11 makes $y^{\text{model}}(\theta; d)$ a standard algebraic expression, so the same autodiff pipeline applies without special support for ODE sensitivity equations.

13 Model-Based Design of Experiments

Model-based design of experiments (MBDoE) selects experimental conditions d that maximize information content of the resulting data, quantified by the Fisher Information Matrix (FIM) $M(\theta, d) = J(\theta, d)^\top \Sigma^{-1} J(\theta, d)$, where $J = \partial y^{\text{model}}/\partial\theta$.[95,94] The `discopt.doe` module implements MBDoE on the same `Experiment` interface used for estimation (Section 12), so the model, responses, and measurement errors are shared across the two steps of a sequential design-then-estimate loop.

Compared with `Pyomo.DoE`,[96] discopt computes J by `jax.jacobian` applied to a parametric DAG-compiled response, giving exact sensitivities in a single forward-mode autodiff pass with no finite-difference step tuning, and supports GPU / TPU acceleration when available through the JAX backend.

Four optimality criteria are supported:

$$\text{D-optimal: } \max_d \log\det M(\theta, d), \qquad \text{A-optimal: } \min_d \operatorname{tr} M(\theta, d)^{-1}, \tag{12}$$

$$\text{E-optimal: } \max_d \lambda_{\min}(M(\theta, d)), \qquad \text{ME-optimal: } \min_d \kappa(M(\theta, d)), \tag{13}$$

evaluated with an optional prior FIM to support sequential designs. `optimal_experiment` performs a multi-start random + boundary search followed by scipy L-BFGS-B local refinement with bound constraints. `sequential_doe` alternates parameter estimation and single-experiment design over multiple rounds, updating the prior FIM with each round's contribution.

Batch / parallel MBDoE. When the laboratory can run multiple experiments in parallel (several reactors, wells, or robots), designing them one at a time wastes information. Because the experiments are independent, the Fisher information is additive,

$$M_{\text{total}}(d_1, \ldots, d_N) = \sum_{i=1}^{N} J(\theta, d_i)^\top \Sigma^{-1} J(\theta, d_i) + M_{\text{prior}}, \tag{14}$$

and the joint D-optimal batch design maximises $\log\det M_{\text{total}}$ over the stacked design vector $(d_1, \ldots, d_N)$. This joint-FIM formulation follows;[30] a modern framework for robust, uncertainty-aware parallel MBDoE is given in.[31] For an alternative route via parallel Bayesian optimization with Gaussian-process surrogates rather than FIM, see.[97]

The `batch_optimal_experiment` function exposes three strategies:

- **Greedy** (default). Pick one design via `optimal_experiment`, fold its FIM into the running prior, repeat. Cheap (N calls to the single-experiment search) and incremental, but myopic: each pick is locally optimal given the previously committed ones and is not guaranteed to recover the joint optimum on nonlinear experiments.
- **Joint**. Stack the N design vectors into $z \in \mathbb{R}^{Nd}$ and optimize the joint criterion directly via scipy L-BFGS-B with a multi-start seeding over random and quantile-structured stacks. Recovers the Galvanin[30] joint optimum up to local-solver convergence; on a two-parameter linear-Gaussian toy ($y = ax + b$, $x \in [0, 1]$) it reliably places $N = 2$ experiments at the boundary support $\{0, 1\}$ where greedy can miss depending on its first pick. Cost scales as $N \times d$ solver dimensions with repeated JAX autodiff passes, so it is slower than greedy for large N but equal or better in information content.
- **Penalized**. Greedy with a normalized minimum-distance filter between successive picks, useful when the criterion is locally flat and the experimenter wants spatial diversity in the batch.

All three strategies are also exposed as a `batch_strategy` option on `sequential_doe` via the `experiments_per_round` keyword, so a batched closed-loop campaign (Estimate → Design batch of N → Run N experiments → Estimate ...) is a single function call. The FIM prior accumulated across rounds ensures the batch criterion evaluated at each round reflects all previously collected data.

13.1 Identifiability and Estimability Diagnostics

Optimal-design criteria assume the model parameters are identifiable in the first place. When the data $\{y_i\}$ and the response Jacobian $J = \partial y^{\text{model}}/\partial\theta$ admit linear dependence between columns of J at the candidate $\hat{\theta}$, the FIM is rank-deficient or near-singular and individual standard errors blow up even though the model fit is excellent. The `discopt.doe` module addresses this with three complementary diagnostics, all built on the same DAG-compiled Jacobian used for FIM evaluation.

Structural identifiability. `compute_fim` returns the noise-weighted FIM and `check_identifiability` reports its numerical rank and null-space directions. A rank-deficient FIM

means that any linear combination of parameters in the null space leaves y^{model} unchanged, i.e. those combinations are mathematically unidentifiable from the response model regardless of how much data is collected.

Estimability ranking. For practically identifiable subsets of an over-parameterised model we implement the orthogonalisation algorithm of Yao et al.[33] in `estimability_rank`, which iteratively selects the parameter whose scaled sensitivity column has the largest residual norm after Gram–Schmidt orthogonalisation against the previously selected columns. The complementary collinearity index γ_K of Brun, Reichert and K{\"u}nsch[34] (`collinearity_index`) measures the worst conditioning of any subset of size K: $\gamma_K = 1/\min_{\|\beta\|=1} \|Z_K\beta\|$ on the Brun-scaled sensitivity matrix $Z = \Sigma^{-1/2} J \operatorname{diag}(s_\theta)$, with the rule of thumb $\gamma_K \leq 10$ for a practically identifiable subset. `d_optimal_subset` dispatches between exact enumeration (small p) and the greedy Yao ranking (larger p) following.[98]

Profile likelihood. For each parameter θ_j in turn, `profile_likelihood` steps θ_j outward from the global estimate $\hat{\theta}_j$, re-solves the estimation problem with θ_j fixed at each step, and tabulates the deviance

$$\Delta D(\theta_j) = D(\theta_j; \hat{\theta}^*_{-j}) - D(\hat{\theta}) \quad \text{vs.} \quad \chi^2_{1,1-\alpha}, \tag{15}$$

following Raue et al.[32] The intersection of the profile with the $\chi^2_{1,0.95}$ threshold gives a likelihood-ratio confidence interval that, unlike the asymptotic Wald interval, **retains its meaning when the FIM is ill-conditioned**: a profile that fails to cross the threshold in one or both directions diagnoses **practical non-identifiability** in that parameter; a flat profile signals **structural non-identifiability**. The deviance convention is fixed throughout (no factor of $\frac{1}{2}$), so the threshold is read directly against `result.objective`.

13.2 Model Discrimination and Selection

Once an experiment campaign has produced data that fits a chosen model, the natural next question is whether a **different** model would explain the data equally well, or whether competing mechanistic hypotheses can be told apart by additional designs. discopt provides a model-discrimination MBDoE module under `discopt.doe.discrimination` and `discopt.doe.selection`, operating in **prediction space**: only $\hat{y}_i(d)$ and the prediction covariance $V_i = J_i M_i^{-1} J_i^\top$ from each candidate model cross model boundaries, so candidates may have entirely different parameter sets without any name alignment.

Five discrimination criteria are exposed via the `DiscriminationCriterion` enum:

- **HR** — Hunter–Reiner,[35] the squared difference of point predictions weighted by inverse measurement variance, $\Phi_{\text{HR}}(d) = \sum_{i<j} (\hat{y}_i - \hat{y}_j)^\top \Sigma^{-1} (\hat{y}_i - \hat{y}_j)$. Cheap; pairwise; ignores prediction uncertainty.

- **BF** — Buzzi-Ferraris–Forzatti–Emig–Hofmann,[36] **the default**, generalising HR to multiresponse models by replacing Σ with $\Sigma + V_i + V_j$, so designs that put predictions far apart are penalised when the predictions are themselves uncertain.

- **JR** — Jensen–R{\'e}nyi divergence between Gaussian predictives $\mathcal{N}(\hat{y}_i, \Sigma + V_i)$ following Olofsson, Deisenroth and Misener.[37] Symmetric, naturally handles $M > 2$ candidates.

- **MI** — mutual information $I(M; y \mid d)$ between the model index and a future observation, estimated by nested Monte Carlo over the Gaussian predictives in the spirit of Lindley[38] and Foster et al.[39]
- **DT** — the DT-compound criterion of Atkinson, Bogacka and Bogacki,[40] $(1-\lambda)\Phi_{\text{precision}}(M_p) + \lambda\Phi_{\text{discrimination}}$, balancing parameter precision (any of the optimality criteria of Section 13 applied to the **preferred** model) against between-model separation. Useful when one model is provisionally preferred but the experimenter wants both to refine it and to keep its rivals in the picture.

The `discriminate_design` entry point picks one design under any of these criteria; `sequential_discrimination` alternates design with parameter re-estimation and updates a running posterior $p(M_k \mid \mathcal{D})$ over the candidate set via Akaike weights[99] or Bayes-factor approximation, terminating either at a posterior threshold (e.g. $\max_k p(M_k \mid \mathcal{D}) > 0.95$) or at a fixed budget of rounds. `model_selection` performs final-round selection by AIC / AIC_c / BIC / cross-validation given the accumulated data.

14 Multi-Objective Optimization

Many engineering problems trade off two or more conflicting objectives — cost vs. recovery, throughput vs. selectivity, capital vs. operating expense — and admit no single best solution but a **Pareto front** of mutually non-dominated alternatives. The `discopt.mo` module provides a scalarisation-based interface for tracing this front.[100,101] The design choice is deliberate: each scalarised subproblem is an ordinary discopt `Model`, so it inherits the full MINLP infrastructure — discrete variables, structural convexity certification (Section 7.6), spatial branch and bound, AMP global solving, the differentiable-solve API of Section 10, and the McCormick / piecewise-McCormick / αBB relaxations. No separate evolutionary or surrogate-based MO machinery is needed.

Given K objectives $f_1, \ldots, f_K$ and a feasible set $\mathcal{X}$, a point $x^* \in \mathcal{X}$ is **Pareto-optimal** if no $x \in \mathcal{X}$ satisfies $f_k(x) \le f_k(x^*)$ for all k with strict inequality for at least one k. The five scalarisations exposed by `discopt.mo` are:

- **Weighted sum** (`weighted_sum`). Minimise $\sum_k w_k g_k(x)$ over a grid of non-negative weights summing to one, with $g_k = (f_k - z_k^*)/(z_k^N - z_k^*)$ if `normalize=True` (the default), where z^* is the ideal point and z^N the nadir point computed from the per-objective payoff table. Weighted sums are computationally cheap and recover the full convex Pareto front, but cannot reach points on a non-convex region of the front regardless of weight choice.
- **Augmented ε-constraint (AUGMECON2)** (`augmecon2`). Minimise $f_1(x) - \delta \sum_{k \ge 2} s_k$ subject to $f_k(x) + s_k = \varepsilon_k$, $s_k \ge 0$ for $k \ge 2$, sweeping ε_k over a grid in $[z_k^*, z_k^N]$ following Mavrotas.[41] The slack variables s_k together with the small augmentation δ guarantee that every solution returned is properly Pareto-optimal (no weakly-dominated points), and AUGMECON2's **bypass coefficient** skips ε grid points that the previous slacks have already shown to be redundant, sharply reducing the number of subproblems on dense grids.

- **Weighted Tchebycheff** (`weighted_tchebycheff`). Minimise $\max_k w_k(f_k(x) - z_k^*)$ via an epigraph reformulation $\min t$ s.t. $w_k(f_k - z_k^*) \le t$. Recovers all Pareto-optimal points (convex **and** non-convex regions), at the cost of a non-smooth max that becomes a single auxiliary variable plus K inequality constraints in the model.

- **Normal Boundary Intersection (NBI)** (`nbi`). Following Das and Dennis,[42] parameterise the front by points on the convex hull of the per-objective anchors and project orthogonally toward the lower-left, producing a uniform spread along the Pareto surface for problems where uniformly-spaced weights produce clustered solutions.

- **Normalised Normal Constraint (NNC)** (`nnc`). The complementary scheme of Messac, Ismail-Yahaya and Mattson,[43] which slides a hyperplane normal across the normalised utopia line, yielding well-spread fronts on smooth bi- and tri-objective problems.

The `ParetoFront` container collects the resulting non-dominated points along with their objective values and original-space variable assignments; `filter_nondominated` post-processes any candidate set into the Pareto subset.

For comparing two front approximations, `discopt.mo.indicators` provides four standard quality measures[44,45] — the **hypervolume** (exact 2-D and 3-D Lebesgue measure of the dominated region behind a reference point, Monte-Carlo for $K \ge 4$), the **inverted generational distance** (mean distance from a reference front to the approximation, capturing both convergence and spread in one number), the **spread** (coefficient of variation of consecutive point distances in objective space, lower is more uniform), and the additive **epsilon indicator** $I_\varepsilon(A, B) = \min \varepsilon$ such that $A \preceq_\varepsilon B$ in the dominance order. All indicators internally convert to the all-minimise convention using the per-objective senses stored on the front, so results are sense-invariant.

Out of scope in the current release and left to future work: evolutionary MO algorithms (NSGA-II, MOEA/D), Bayesian MO acquisition functions over Gaussian-process surrogates, and interactive / preference-articulation methods that elicit the decision-maker's value function during the search rather than offering a static front.

15 Certificate Soundness

A global solver's product is not a number but a *certificate*: a claim that the returned point is globally optimal (or that the model is infeasible) backed by a valid dual bound. A change that makes the solver faster but admits a false `optimal`, false `infeasible`, or invalid bound on any instance is a regression, not an improvement. We therefore treat soundness as a first-class deliverable and report a systematic audit of the certificate, conducted as roughly three dozen tracked correctness issues (each with a deterministic reproducer, a fix, and a regression test) together with a twenty-module code review of the whole codebase. The governing invariant throughout is =incorrect_count = 0= on the certifying panel — a hard gate with zero slack that no fix is permitted to weaken.

The issues cluster in the layers where an unsound step silently corrupts the bound:

- **Branch-and-bound status logic.** A nonconvex model whose root cannot be spatially branched (a variable with a one-sided or fully free domain) and for which no relaxation ever established a finite dual bound was falsely certified `optimal` with zero gap on a local incumbent, because the tree collapsed the global lower bound onto the incumbent. The fix makes the tree carry the unresolved-bound condition explicitly, so such a root reports `feasible`*unknown* — *it /abstains* from the optimality claim — and never a false optimal. Related fixes require a rigorous proof (not a mere NLP-solve failure) before fathoming a node infeasible, and refuse to trust an under-converged NLP objective as a node lower bound.

- **Relaxation math.** Several McCormick envelopes were made sound on their domain edges: `relax_tan` now abstains to the no-information envelope across a pole rather than drawing a secant through it; `relax_div` falls back to a sound interval enclosure for a nonlinear denominator instead of an invalid convex underestimator; `relax_asin` / `relax_acos` use the correct curvature regime; and infinite-bound boxes degrade to the sound no-information bracket rather than producing `NaN`. The αBB node bound was routed through a rigorous per-box α and now abstains whenever a rigorous α cannot be certified, replacing an unsound sampled-α / centre-only positive-semidefinite check.

- **Presolve and FBBT.** Feasibility-based bound tightening on an array-valued variable block previously collapsed the block to a single element's bounds and stamped them on every element, which could both cut feasible points and report false infeasibility; it now seeds each block from the element-wise *union* of bounds, a valid outer box. A presolve bound-resync after a variable-removing pass no longer fuses the bounds of unrelated variables. Interval multiplication that produced `NaN` on the $0 \cdot \infty$ corner (a lost-tightening bug, never an unsound verdict) was closed on both the Rust and Python paths.

- **Cuts and disjunctions.** A no-good / outer-approximation cut is emitted only on rigorously proven infeasibility; Gomory-style cuts carry a right-hand-side safety margin; and a Farkas ray is verified before an LP-infeasible fathom. In the generalized-disjunctive layer, a big-M reduction that treated a large-but-finite default bound as effectively infinite and substituted a far-too-small M — turning an inactive disjunct's constraint into a real one and producing a false `infeasible` on the *default* solve path — was corrected to use a finite bound as a valid M and to refuse loudly when no finite M exists. An even-power bound over a zero-straddling base now includes the interior minimum.

- **The `.nl` parser and interop.** The MINLPLib `.nl` import path was hardened to inline defined variables, handle additional opcodes, preserve integer and binary variable bounds, and refuse (rather than silently misparse) unsupported `floor`/`ceil`/`round` constructs; the Jacobian sparsity structure was made ASL-conformant.

- **The LLM front-end.** An expression-evaluation path in the (optional) natural-language model-builder that used Python `eval` was replaced with a whitelisted-abstract-syntax-tree evaluator, closing a remote-code-execution class while preserving the safety invariant that LLM outputs never affect solver math.

The audit also produced affirmative *soundness confirmations* where the review found no defect: the structural convexity certifier and convex fast path — the highest-stakes gate, since a wrong convex verdict would certify a nonconvex model's local optimum as global — were verified sound under a nine-hundred-plus-certificate randomized fuzz with zero false verdicts, as were the primal/dual simplex, the branch-and-bound tree manager, the PyO3 boundary, and the RLT/PSD/SOC/edge-concave cut families. The recurring discipline across every fix is the same: where a rigorous guarantee cannot be established, the solver *abstains* (returns a weaker but valid status or bound) rather than emitting an unsound certificate.

16 Saved Models, Saved Results, and Provenance

A solved model is a research artifact, and the point of saving one is to come back to it — or to hand it to a reviewer — and reproduce the number. discopt therefore ships a native round-trippable format, `.dopt`, alongside the exchange formats it reads. The exchange formats lose exactly what matters for that purpose: `.nl` renames every variable positionally, neither `.nl` nor `.gms` carries the fitted solution, and both refuse the non-algebraic relations — indicator, SOS, disjunction — outright.

The format is a single JSON document, optionally gzipped, and the choice over `.npz` or pickle was deliberate: JSON is readable from any language, carries no numpy format version and no Python version, and is greppable and diffable in git; float text compresses well enough that gzip covers the size concern without a binary container. The expression DAG is written as a flat node table rather than a nested tree, because expressions share subnodes heavily — a collocation model with one surrogate reuses the same subexpression at every collocation point — and nesting would re-expand every share. The table's array positions *are* the node ids, so sharing round-trips exactly with no separate identifier bookkeeping.

What the document carries is stated exhaustively, and so is what it refuses. It carries variables with element-wise bounds, parameters, the objective, algebraic constraints, the Rust-builder linear blocks, indicator, SOS and disjunctive relations, the complementarity relations *together with this model's lowering marks*, the starting point, optionally the solve result, and — from this release — the registered-atom tags of Section 7.2. It refuses, loudly and by name, an opaque `CustomCall` (an arbitrary Python callable: there is nothing faithful to write, and writing its *name* would reload into a different model), a propositional logical constraint, a fancy array index, and any node this version does not know. On read, an unknown opcode is a hard error, because a reader that skips what it does not understand silently drops a constraint and returns a model that solves to the wrong answer.

Two details are worth stating because each closes a way of being silently wrong. First, the complementarity state travels in *both* halves. Carrying the relations without their lowering marks reloads a model whose rows already encode every relation but which reports them all unlowered — and since an unlowered relation is a solver-boundary refusal (Section 4.7), the reloaded model would not solve at all. Carrying the marks without the *method* is worse: exactness is derived from the method alone, a Scholtes lowering is a relaxation, and a mark that lost its method reads as exact — so the solver would certify a relaxation as though it were the declared model. Second, the schema minor is a real version and is read, not decoration. A minor moves whenever absence of a key acquires a second meaning: the registered-atom

tag is additive — a reader that ignores it gets the lowering and relaxes it term by term, which is sound — yet the minor still moves, because "this model has no registered atoms" and "the writer could not record them" are different facts and only the minor distinguishes them. A differing *major* is refused rather than guessed at.

The provenance block records who wrote the document, when, and with what: a UTC timestamp; the software identity including the version of the compiled Rust core, which holds the expression IR, the `.nl` parser and the LP layer and can move independently of the Python version (it is the half that `pip show` does not report); the git commit `HEAD` points at, read directly from `.git` with no subprocess, because `git rev-parse` on a large or network-mounted repository can take hundreds of milliseconds and this runs on every save; a source fingerprint combining the version with the newest modification time across the installed package tree; and the interpreter, implementation, system and machine, because a result can depend on all four through floating-point details. The commit and the fingerprint are an honest pair where either alone would not be — the commit cannot see uncommitted changes (detecting them needs the index comparison the subprocess-free read exists to avoid), and the fingerprint is exactly what moves when an editable checkout is edited without committing.

A document loaded and re-saved keeps its predecessor's block under `derived_from`, but **exactly one level deep**: the predecessor's own `derived_from` is dropped, so a model round-tripped a thousand times does not carry a thousand-deep chain. That is a real loss of the deeper history, and it is stated in the docstring rather than left to be discovered — the full chain lives in the files themselves, each retaining its own immediate predecessor.

The author field is optional and **never inferred**. It is recorded only when the caller passes one or sets the corresponding environment variable, and is deliberately not read from the repository's citation file: that file names the author of *discopt*, while a saved model is the user's artifact, and copying discopt's author onto a stranger's model would be a false attribution — precisely the class of metadata error FAIR provenance exists to prevent.

It should be stated plainly what this is and is not. **Provenance here is a record, not a reproducibility guarantee.** Nothing in the block is read back into the model's mathematics: the reader attaches it to `Model.provenance` and warns on a writer/reader version skew, and that is the whole of its effect. The warning has its own category, so a pipeline that requires an exact version can escalate it to an error. The claim that no content of the block reaches the mathematics is asserted directly rather than rhetorically — `loads` is fed a corrupt provenance block carrying keys like `objective` and `variables`, and the rebuilt document is required to be byte-identical. Recording provenance is on by default and can be turned off for a byte-reproducible document, since the block carries a timestamp; the docstring names that for what it is, a deliberate loss of provenance rather than a default.

The same rule governs saved *results*, where it has a sharper edge. **The provenance of a result is the identity of the process that solved it**, which on the daemon path is not the process that writes the file: a warm daemon serves the solve in its own interpreter, so the daemon stamps the block and the client carries it to disk unchanged. Recomputing it at the writer would record the client's version for a number the daemon computed — quietly wrong in exactly the way provenance exists to prevent. Hence a carried value always beats a freshly captured one, and requesting provenance means "stamp one if this result does not already have one".

A result file also carries the validation report. For a solver whose product is its certificate,

an archived result that silently omits the validation it passed is a worse gap than a missing timestamp. The report is the Examiner-style audit of Section 15 — re-evaluate the model at the returned point, recover multipliers from the active set when the solver does not expose them, and report the first-order KKT residuals together with integrality and objective consistency — and it round-trips as the real dataclass rather than as a dictionary, because on the daemon path the client deserializes the reply and then writes the file. An object that is not such a report raises rather than being dropped, since a report that silently vanishes from an archived result is the failure this change exists to fix. The infeasibility certificate travels too, with one caveat recorded in its docstring: its violation arrays are indexed in backend LP row order, not in the order of the constraints the user wrote, and the file does not carry that mapping.

Finally, every float the writer emits — scalars, solution and dual arrays, and the nested blocks — goes through tagged encoding, so a NaN or an infinity is written as a tagged value rather than as the bare `NaN` / `Infinity` tokens that Python's `json` emits and other languages' parsers reject. The document is dumped with `allow_nan=False`, so a float that slips past the encoders raises instead of quietly producing a file that is not JSON. Where the reader knows a position holds a number a bare token suffices; the free-form option and trace subtrees, which may hold a string of any text, use a structurally tagged form instead, since a string option reading "nan" could not otherwise be told from the number.

17 Computational Experiments

17.1 NLP Backend Comparison

This comparison is the measurement that decided the architecture, and we report it in the configuration in which it was taken — when all three backends still existed — rather than re-running it on a tree from which one has been removed. Five test problems span unconstrained quadratics to constrained portfolio optimization; Table 3 reports objectives and wall-clock timing, collected on an Apple M4 Pro (CPU backend, JAX 0.8.2), as single-solve wall-clock time including model compilation. The `IPM` column is the pure-JAX IPM of Section 5.1, which has since been retired; POUNCE and Ipopt are the two backends that remain.

Problem	f^*	IPM (s)	POUNCE (s)	Ipopt (s)
Quadratic ($n = 2$)	0.000000	0.023	0.033	0.013
Constrained ($n = 5$)	1.250000	0.269	0.001	0.001
Rosenbrock ($n = 2$)	0.000000	0.590	0.082	0.114
Sum-of-squares ($n = 10$)	0.000000	1.189	0.049	0.070
Portfolio ($n = 8$)	0.002064	0.229	0.004	0.001
Total		2.30	0.17	0.20

Table 3: NLP backend comparison. All backends achieve identical objectives (to solver tolerance). Timings on Apple M4 Pro, CPU.

Key observations:

- All three backends produce **identical optimal objectives** to solver tolerance (10^{-6}), confirming correctness across the Rust, JAX, and C++ implementations.
- POUNCE is competitive with Ipopt (total 0.17s vs 0.20s), and faster on Rosenbrock (0.082s vs 0.114s) and sum-of-squares (0.049s vs 0.070s).
- The pure-JAX IPM includes JIT compilation overhead on first evaluation; subsequent solves of similarly-structured problems benefit from cached JIT traces. Its total, 2.30~s, is an order of magnitude above both alternatives, and the cold-start component does not amortize in a spatial B&B whose nodes keep changing shape.
- On a quartic constrained NLP ($n = 10$), POUNCE (0.334s) and the JAX IPM (0.795s) agree to 6.5×10^{-15} in objective value, confirming numerical equivalence.

Read together, these are a retirement notice. The JAX IPM was correct — agreement with Ipopt to 10^{-15} is as strong a statement as this panel can make — and it was uniformly the slowest, while POUNCE delivered the same algorithm at Ipopt's speed with no C or Fortran dependency. Once a Rust implementation existed, the JAX one bought only differentiability, and differentiability is supplied more cheaply by implicit differentiation of the KKT system at the solution (Section 10) than by making the whole solve traceable. It was removed, along with the JAX LP and QP IPMs, and the derivative evaluator followed; Section 18 reports what that cost and what it bought.

17.2 MINLP Correctness

We validate on 24 MINLP problems (12 NLP + 12 MINLP with binary/integer variables) from the discopt test suite, with known optimal values. This panel, together with the vendored MINLPLib instances below, forms the *certifying panel* on which the =incorrect_count = 0= gate of Section 15 is enforced — every soundness fix in the audit is required to hold this gate:

- **24/24 solved correctly** with zero incorrect results
- Tolerances: absolute 10^{-4}, relative 10^{-3} for objectives; integrality tolerance 10^{-5}
- Problems include Rosenbrock, exponential, logarithmic, trigonometric, bilinear, and cardinality-constrained formulations

B&B convergence with the POUNCE backend was verified on two MINLP instances: a simple 3-variable problem (optimal $f^* = 2.0$, 1 node, 0.039s) and a larger 6-variable formulation (optimal $f^* = 3.0$, 13 nodes, 2.38s).

17.2.1 MINLPLib Validation

We further validate on instances from MINLPLib,[83] imported via the `.nl` file parser. Table 6 reports results on a representative 9-instance smoke test suite. (This early smoke test predates the pure-Rust LP core of Section 6 and the relaxation-coverage work of Section 7, which certify many more of these instances; the head-to-head below is the current measurement.)

Head-to-head against BARON and SCIP. We compare discopt against two state-of-the-art global solvers — full-license BARON[1] run through GAMS and SCIP[2] — under a matched 60~s per-instance budget and 10^{-4} relative gap tolerance, taking the MINLPLib primal bound as the oracle. Two panels are reported, because the two baselines cannot be driven through one harness: SCIP reads `.nl` directly and runs in-process, while the only full BARON licence available to us lives inside GAMS and reads `.gms`. The wide panel (Table 4, 199 instances spanning convex and nonconvex MIQP, nonconvex MINLP and nonconvex NLP) is therefore discopt against BARON; the three-way panel (Table 5, 50 instances) adds SCIP. Both tables are produced by `discopt_benchmarks/scripts/panel_tally.py` from the archived results JSON named in each caption, which re-derives every verdict through the harness's classifier rather than replaying the verdict stored at run time.

Two columns are reported separately throughout, and the distinction is the whole point. *Certified* counts the instances on which a solver **asserted** a global certificate. *Matches oracle* counts the instances whose returned objective agrees with the published optimum to tolerance, whether or not the solver proved it. A solver can match the oracle without certifying — it found the answer but did not close the gap — and the two numbers are not interchangeable.

Solver	Certified	Matches oracle	Gap	Violations	No incumbent	Median (s)	Total wall (s)
discopt	121	158	20	0	21	2.38	4564.7
BARON	97	166	27	1	5	0.05	919.7

Table 4: discopt vs. full-licence BARON on 199 MINLPLib instances, 60~s per instance, Apple M4 Pro. From `reports/global_opt_baron_vs_discopt_2026-07-18T14-09-20.json` via `panel_tally.py`. Median is over oracle-matched instances; total wall is over all 199.

Solver	Certified	Matches oracle	Gap	Violations	No incumbent	Median (s)	Total wall (s)
discopt	48	48	0	0	2	0.33	294.7
BARON	45	49	0	1	0	0.04	130.7
SCIP	44	48	1	0	1	0.06	390.3

Table 5: Three-way comparison on the 50-instance `global50` panel, 60~s per instance, Apple M4 Pro. From `reports/global50_3way_2026-07-28T08-03-28.json` via `panel_tally.py`.

Soundness. discopt records **zero correctness violations** on either panel: no run certified a global optimum at the wrong value, returned an incumbent strictly better than the proven optimum, or reported a dual bound crossing the oracle. Every instance on which discopt claims a certified global matches the oracle to the 10^{-4} tolerance. This is the deliverable the certificate audit of Section 15 exists to protect, and it holds against decades-tuned commercial and academic baselines alike. The two violations recorded against BARON are reported for completeness and we do not build an argument on them: `ex1252`, where GAMS returns `1 Optimal` at 223191.34 against a published 128893.74, and `hda`, where BARON's

preprocessing asserts infeasibility on an instance with a published finite optimum — the same instance is one of discopt's two misses on the three-way panel, and disabling BARON's bound tightening recovers the published value, so it is a tightening bug rather than a search failure. The verdict rule is solver-agnostic by construction: discopt's own `infeasible` would be judged identically.

Coverage, and a correction. Earlier drafts of this section reported that BARON and SCIP each certify on the order of 180 of the ≈ 200 instances against discopt's ≈ 120, and concluded that discopt trails on coverage. That comparison was not like-for-like: the ≈ 120 figure is discopt's **certified** count (121, which the artifact confirms) while the ≈ 180 figure was BARON's **oracle-matched** count (166). Set side by side on the same axis, discopt certifies more of the wide panel than BARON's GAMS status channel reports (121 vs 97) and matches the oracle on slightly fewer (158 vs 166); on the three-way panel the certified counts are 48, 45 and 44. We draw no superiority claim from the certified column in the other direction either, because BARON's certificate cannot be read reliably from the GAMS return code — 88 of the 199 rows come back `2 Locally Optimal` from a solver that is globally convergent, so the status channel undercounts what BARON actually proved. The defensible statement from these artifacts is that the three solvers are **close on coverage**, and that no ranking between them survives the ambiguity in BARON's status channel.

Where discopt genuinely trails: time. The gap that the artifacts do support unambiguously is wall-clock. On the wide panel BARON consumes 919.7~s of total wall against discopt's 4564.7~s, a factor of 5, and its median over oracle-matched instances is 0.05~s against discopt's 2.38~s — nearly two orders of magnitude. The three-way panel shows the same ordering at smaller scale. We report both statistics because a solved-only median flatters whichever solver times out most, its slowest instances having left the population; the total-wall column is the one that cannot be gamed this way. The shortfall concentrates on harder medium-sized nonconvex instances (the `casctanks`, `heatexch_gen*`, `contvar` and `tls2` families among them), where a serial per-node engine cannot close the optimality gap in 60~s. This is a per-node throughput limitation, not a correctness or relaxation-validity one — Section 18 reports where that per-node time is actually spent — and it is the principal open performance work, tracked as issues #727, #723 and #282.

The honest headline is therefore: discopt's certificate is **sound everywhere it is exercised**, its coverage is **comparable to** BARON's and SCIP's on these panels rather than clearly behind them, and it **trails on wall-clock by roughly a factor of five in aggregate** — a per-node throughput gap on a serial engine, not a soundness or bound-validity deficit.

All 9 instances now certify global optimality, and every certified objective matches the MINLPLib oracle to the 10^{-3} reporting tolerance; the notebook that produces this table asserts the comparison on all 9 rows and reports the executed comparison count, so a silently-skipped check cannot read as a pass. Every instance closes in well under one second, against a 60~s budget.

Earlier revisions of this table were taken before the `.nl` import path carried the relaxation and dual-bound machinery of Sections 7 and 7.5. At that point `gear` and `nvs01` stopped at a feasible point (`nvs01` at a local optimum of the wrong sign), and `chance` was reported as optimal at 29.421 against a known optimum of 29.894. Those numbers are superseded by the run above — the node counts in this table are single-digit to low-tens where they

Instance	n	m	Status	f^*	Best Known	Time (s)	Nodes
chance	4	3	optimal	29.894	29.894	0.016	0
dispatch	4	2	optimal	3155.288	3155.288	0.215	23
ex1221	5	5	optimal	7.667	7.667	0.262	5
ex1226	5	5	optimal	-17.000	-17.000	1.072	5
gear	4	0	optimal	0.000	0.000	0.039	3
nvs01	3	3	optimal	12.470	12.470	0.726	3
nvs03	2	2	optimal	16.000	16.000	0.006	0
nvs04	2	0	optimal	0.720	0.720	0.045	5
nvs06	2	0	optimal	1.770	1.770	0.160	5

Table 6: MINLPLib smoke test results (9 instances, 60s time limit, v0.9.0, Apple M4 Pro). Best known objectives from MINLPLib. Reproduced by the manuscript smoke-test notebook.

were previously in the thousands, which is the relaxation doing the work rather than enumeration. We have separately validated the expanded path head-to-head against BARON[1] on the subset of MINLPLib instances with a proven global optimum, matching the known optimum (to the 10^{-4} correctness tolerance) with zero incorrect results; the residual cases where discopt finds the optimum but does not yet prove it are concentrated in high-degree factorable products whose lifted relaxation is loose, consistent with the limitation noted in the Discussion.

17.2.2 The QPLIB Corpus: An Independent, Point-Verifiable Oracle

MINLPLib exercises general factorable nonlinearity. QPLIB[102] — 453 quadratic instances, 390 of them nonconvex — concentrates instead on nonconvex quadratic objectives and constraints, which is exactly the structure the McCormick and αBB layer of Section 7 is built to bound. discopt reads the native `.qplib` format directly, with no AMPL or GAMS conversion step, and the library has no overlap with the MINLPLib snapshot.

The reason for ingesting it is **oracle independence**, and it is a correctness argument rather than a coverage one. The quadratic coverage discopt already had consisted of small synthetic families whose optima *we* computed ourselves by vertex enumeration and multistart — an oracle only as good as our own code, which is the wrong property for a correctness corpus. More importantly, QPLIB ships reference solution *vectors* and not merely published objective values. A published objective is one number produced by someone else's model: matching it establishes agreement on that number, not that our parsed model *is* their model. A reference point closes both loops at once — feasibility of x against our own parsed rows, bounds and integrality, and reproduction of the file's own objective value at that point.

For this format the distinction is load-bearing rather than philosophical. The `.qplib` layout carries no section markers and is conditional on a three-character problem-type code: whether the objective-quadratic block, the variable bound block, the variable type block, the constraint count line and the constraint-quadratic block are present at all depends on that code. Getting the conditioning wrong **does not raise** — it shifts the token stream and

yields a well-formed but wrong model, which is precisely the failure a vector-based check catches and a scalar check does not. Two undocumented upstream conventions were in fact recovered *from the reference points* rather than from documentation. The first is the uniform factor of $\frac{1}{2}$ on every stored quadratic entry, off-diagonal as well as diagonal: only the lower triangle is stored, and a stored off-diagonal entry represents the combined $Q_{ij} + Q_{ji}$, so it must not be doubled on symmetrization. The tempting reading — halve the diagonal, take the off-diagonal at face value — overstates every off-diagonal contribution by a factor of two and yields objectives roughly twice the published ones. A least-squares fit for the two coefficients over 40 all-continuous instances (where $x^2 \neq x$, so the diagonal is identifiable) recovers $(\frac{1}{2}, \frac{1}{2})$ at rank 2 with a maximum residual of 4.5×10^{-12}, and that fit is re-run as a unit test so a silent convention change cannot pass. The second is the solution-file index offset: record names are GAMS variable numbers counting the objective variable as variable 1, and the adopted offset yields zero out-of-range and zero type-mismatched records against 93 and 34 respectively at the naive offset. Neither convention is derivable from objective values alone.

Instance selection is by manifest rather than by name. A generated manifest records, per instance, its problem type, sizes, integrality counts, convexity, nonzero structure, the reference value, whether the reader reproduces that value at the reference point, and the maximum violation of that point in the parsed model; `usable_oracle` is the conjunction of the last three. **421 of the 453 instances qualify.** The 32 that do not are accounted for exactly: 27 ship no reference point at all, 4 are upstream-defective, and 1 has a reference point that is marginally infeasible at a violation of 2.4×10^{-4}. The four defective instances are excluded by a *general rule* rather than by a name list — the upstream metadata counts the variables appearing in the objective, so a positive count with nothing stored in the file means the objective was lost upstream — and their reference points are grossly infeasible besides, so they cannot serve as oracles under any reading. A future upstream release that repairs them needs no code change, only a new expected count in the test.

The instrument behind that count carries its own scar. The manifest recomputes `usable_oracle` by truthiness and **errors out** if the recount disagrees with the identity count, because numpy returns `np.bool_` and `np.bool_(True) is True` evaluates false — which silently sank the usable-oracle count from 421 to 216 the first time the manifest ran. The reference-point reader is called deliberately unguarded, since a malformed reference point is a reader defect and hiding it would make the manifest meaningless.

All 453 instances parse, and 4448 assertions against the library's own metadata and reference points pass, with failures confined to the five instances documented above. Nine instances are vendored as fixtures covering every conditional layout branch small enough to vendor and run in CI in under a second; the full-corpus check covers the remaining branches.

What we do not claim. QPLIB is ingested and verified, but it is **not yet wired into a benchmark suite**, and this paper reports **no QPLIB solve panel**. What the corpus has supported so far is relaxation-level measurement. The clearest of these is the root-bound effect of the linear-equality (Sherali–Adams constraint-factor) RLT family[46] on continuous nonconvex QPs: for a linear equality $a^\top x + c = 0$ and a variable x_j, the product $(a^\top x + c)\, x_j = 0$ linearizes over the lifted product columns to $\sum_i a_i X_{ij} + c\, x_j = 0$, which holds with equality at every feasible point and so can never cut one, and which — unlike the bound-factor families — contains no box data and is therefore identical at every node.

Table 7 reports the measured root LP bound with and without it. The McCormick-only root bound is hopeless on this class, and on the instances that respond, *all* of the recoverable gap comes from this equality family; the bound-factor rows added nothing on top of it. No bound crossed its published optimum on any instance in the probe.

Instance	McCormick	+ linear-eq RLT	Optimum	Gap closed
QPLIB_1157	-14.8046	-11.7716	-10.9482	78.6\%
QPLIB_1493	-85.6492	-65.5915	-43.1604	47.2\%
QPLIB_1143	-140.3576	-114.7060	-57.2467	30.9\%
QPLIB_1423	-25.6777	-22.6093	-14.9675	28.6\%
QPLIB_1507	-18.1057	-15.8070	-8.3014	23.4\%

Table 7: Root LP bound on five continuous nonconvex QPLIB QPs: McCormick alone, plus the linear-equality RLT family, against the published optimum. The family is gated `off` by default pending the graduation panel of Section 15.

That family is **default-off**, and the policy of Section 15 is why: a bound-changing feature graduates only on a corpus-wide differential panel that is both certificate-clean and measurably net-positive, and a root-gap improvement on five instances is an entry measurement, not that panel. The same corpus has also produced a cautionary cold-start measurement in the other direction — on these instances warm-starting the relaxation LP cold costs between 3× and an iteration-cap failure — which is exactly the kind of result a five-instance root probe cannot anticipate.

17.3 Neural Network Embedding Validation

We validate the `discopt.ml` module on feedforward networks with ReLU and smooth activations. A representative 2-input, 4-hidden, 1-output network demonstrates the three formulation strategies:

- **ReLU big-M**: 4 binary variables (one per hidden ReLU neuron), 9 continuous variables, 16 linear constraints. Interval arithmetic propagates tight pre-activation bounds, simplifying neurons whose bounds are entirely positive or negative to linear constraints with no binary variable.
- **Full-space** (tanh): 9 continuous variables, 8 nonlinear activation constraints. McCormick relaxations applied to tanh activation functions provide valid lower bounds within spatial B&B.
- **Reduced-space**: zero intermediate variables; nested expressions build a single composite function, suitable for networks with ≤ 3 layers.

All three strategies produce identical optimal objectives on the test instance, confirming correctness of the algebraic encoding. The ReLU big-M formulation solves to global optimality at the root node (no branching required) when pre-activation bounds are tight enough to fix all binary variables; the full-space formulation requires 1–3 B&B nodes with McCormick

relaxations. The ONNX reader is validated against PyTorch-exported models: imported weights and biases match to machine precision, and `NetworkDefinition.forward()` reproduces the PyTorch output to 10^{-14} absolute error.

17.4 Relaxation Gap Reduction

Table 2 reports the envelope-gap reduction of piecewise McCormick, the relaxation AMP refines. Additionally:

- α BB node bounds use only the rigorous interval-Gershgorin α. A randomized differential panel of up to 100 sub-boxes of a nonconvex objective with a narrow negative-curvature spike (the shape that defeated the former sampled estimator) finds no bound above the dense-grid box minimum, and the routine abstains rather than bound where the interval Hessian is unbounded (`python/tests/test_alphabb_bound_soundness.py`).
- The alternative bounding route that offered `piecewise`, `learned` (ICNN), and Taylor/Chebyshev/ellipsoidal arithmetic modes was measured before it was retired. It could fire on only 7 of the 66 in-repo corpus instances, and only under a non-default option combination; there, ablating its bound changed status, bound, and node count in none of 36 instance/variant pairs. Single, non-interleaved runs (indicative, not a timing claim) put its cost at 1.4–3.3× wall and 2.3–3.6× peak memory on the terminating instances; on two further convex instances it ran 27× over its time limit and was killed out of memory at 13.6~GB, where the default certified in seconds. The learned mode had never taken effect, because no trained relaxation models were ever shipped. Those options are now refused with an error that points at the measurement (`docs/dev/nlp-mccormick-route-retirement.md`).

17.5 Performance

Algebraic coefficient extraction. For LP and QP subproblems, walking the expression DAG directly to extract coefficient matrices avoids expression-level autodiff altogether. Measured against the then-current generic path, which traced through JAX:

- LP ($n = 100$): 3.5s → 0.20s warm (17× speedup)
- QP ($n = 100$): 65.7s → 0.29s warm (226× speedup)

Batch evaluation. For batch size 512 with $n_{\text{vars}} = 50$, `jax.vmap` achieved approximately 29.6× speedup over sequential evaluation — the result that motivated batched node dispatch in the first place, and the reason the batch interface was kept when the backend beneath it moved to Rust.

Interop overhead. Rust ↔ Python array transfer via PyO3/numpy: sub-microsecond for small batches, 0.03–0.14% overhead relative to solver time (well under the 5% performance gate).

Time breakdown. An earlier revision of this paper reported that JAX numerical computation dominated at 92–99.9% of wall time, with Rust tree management at 0.05–7.3% and Python orchestration at 0.02–0.6%. That measurement is **superseded and retracted**: it described a build in which every nonlinear solve routed its derivatives through the JAX evaluator, and the default path no longer does. The current account is a two-arm profile over the whole in-repo corpus (66 `.nl` instances, 20~s budget each, 5185 nodes, 504.9~s of unprofiled wall, 333 executed assertions). A *clean* arm takes wall, nodes, and the FFI-boundary split with no profiler attached; a *cProfile* arm re-runs the same solves and classifies every entry by the layer its code lives in, aggregating by **self** time so a native frame that calls back into Python is not credited with the callback. Shares are wall-weighted over the corpus, with the per-instance median beside them so that one long instance cannot carry the table:

- POUNCE native (IPM and AD tape): 47.7% (median 43.9%, range 0.0–79.0)
- `discopt` Python: 15.4% (median 19.2%, range 2.5–55.0)
- `discopt._rust` (LP/MILP): 15.0% (median 4.8%, range 0.2–54.5)
- other Python (stdlib, builtins): 9.4% (median 10.6%)
- evaluator callback glue (Python): 9.2% (median 7.5%)
- numpy/scipy Python: 3.4% (median 4.9%)
- jax / XLA: 0.00% (median 0.0%, range 0.0–0.0)

The attribution has *inverted* relative to the retracted table. The dominant cost is the node and root NLP solve — `solve_nlp` accounts for 57.4% of profiled wall, and 44.9% of all self time is POUNCE's own interior-point iteration, code with no Python in it — while the Python marshaling that the old table put at 82.5% is 23% and falling. The Rust LP is the second consumer at 9.3%. JAX is absent by measurement rather than by assumption: the probe carries a control arm under `DISCOPT_NLP_EVAL=jax` that *does* import JAX (217 modules in `sys.modules`) and *does* pay for it, so an instrument that simply could not see imports would have been caught rather than read as a pass.

Two limitations are part of the result. First, cProfile distortion is not negligible: profiled wall over clean wall is median 1.23× (0.63–2.18×), and 8 of the 66 instances land on a different node count under the profiler. Python-frame-heavy instances inflate most, so every Python share above is an **upper** bound and every native share a **lower** bound — the direction of the headline is therefore safe, because native dominance is understated rather than overstated. Second, this panel was taken on a quiet four-core Linux container, not on the Apple M4 Pro used elsewhere in this section; a calibration probe puts that box at 15.1× slower on an `alan` solve, so what transports across machines is the *composition* reported here and not the absolute times, and no wall-clock comparison against another solver is drawn from it. The claim that the multi-language architecture introduces negligible *interop* overhead is unchanged and separately measured above; what is corrected is which layer the work is in.

Where JAX is, and is not. The derivative evaluator on the default path is an AD tape built by POUNCE, which supplies f, ∇f, g, J, and the Lagrangian Hessian with no JAX at all. It graduated to default-ON under the flag discipline of Section 15 after agreeing with the JAX evaluator over 66 corpus instances to 5.5×10^{-16} (f), 3.8×10^{-16} (∇f), 3.1×10^{-15} (J), and 7.8×10^{-13} (Lagrangian Hessian) — against pre-registered bars of 10^{-10} on the gradient and Jacobian and 10^{-8} on the Hessian. The JAX evaluator remains in the tree behind `DISCOPT_NLP_EVAL=jax`. Measuring that opt-out arm on the ten-instance v0.8.0 panel that preceded the corpus profile above puts the JAX share at 3.4–12.9%: an order of magnitude below the retracted figure *even with JAX switched on*, because the dominant cost was never the derivative evaluation. Node counts are identical between the two arms on nine of the ten instances (`tanksize` differs, 17,095 against 17,203, which is why swapping AD engines was treated as bound-changing rather than as a refactor), and the tape arm is faster on every instance that loads JAX at all.

JAX-free linear and quadratic paths. Independently of the evaluator default, the LP, MILP, QP, and MIQP paths import no relaxation machinery at all: the factorable relaxation, NLP evaluator, and αBB modules are loaded lazily and only when a genuinely nonlinear relaxation is built, and MIQP node relaxations are solved as quadratic programs through POUNCE. A purely linear or quadratic model therefore pays none of the cold-start cost (∼0.5–1∼s of import and first-trace compilation) that the optional differentiable path still requires.

Simplex pivoting. The Rust dual simplex used for LP and MILP node relaxations replaces Dantzig pricing with dual Devex reference-weight pricing,[103] a steepest-edge approximation that reduces the dual iteration count, and replaces the single-breakpoint dual ratio test with a bound-flipping (long-step) ratio test that resolves more infeasibility per pivot by flipping finite-bounded nonbasic variables to their opposite bound as it walks the breakpoints. Both are standard ingredients of production simplex codes and are paired with the LP equilibration and shared-factorization reuse described earlier to keep the LU well-conditioned across a node's strong-branching probes.

Per-node subsolver: re-profile and a bound-neutral lever. A re-profile of the spatial-B&B nodes on nonconvex instances identified the *per-node subsolver* — not tree management or Python orchestration — as the dominant wall-clock cost, and localised it to the LP solves issued by the cut separators and by strong branching. On the `nvs17` instance these families issued on the order of 240 cold interior-point LP solves (≈ 40% of wall time) that the in-house warm-started Rust simplex resolves in a few milliseconds each. Routing both LP families to the warm simplex is a *bound-neutral* change — the separator consumes only the LP's dual slope and recomputes the cut intercept to the exact validity boundary, and strong branching reads only the child objective bound, never the vertex — and it was verified neutral on the certification baseline: node counts were *exactly* unchanged across the panel, objectives bit-for-bit identical, and =incorrect_count = 0=. The measured effect is that `nvs17` moves from a 60∼s time-limit miss (feasible) to a certified optimum in 38∼s (per-node time roughly halved, ≈ 2×), and a companion instance `nvs13` from 3.9∼s to 1.4∼s (≈ 2.7×); instances that issue no separation or strong-branch LPs are unaffected. The residual per-node cost was traced to the incumbent NLP subsolves: an apples-to-apples replay of the same subproblems from the same starting points found the in-house POUNCE backend roughly an order of magnitude slower than Ipopt (pooled median ≈ 7×; ≈ 11× on

the subset reaching the same optimum), a POUNCE performance problem with a minimal reproducer that has been filed for upstream work rather than papered over.

18 Discussion

discopt demonstrates that a modern, multi-language MINLP solver can provide capabilities absent from existing monolithic solvers: differentiability, batch evaluation, and memory safety. Several limitations and directions for future work deserve discussion.

Limitations. POUNCE uses dense linear algebra, which limits scalability to problems with $n \lesssim 5000$ variables; for larger problems Ipopt, with its mature sparse factorizations, should be used. The `.nl` import path now constructs convex relaxations and surfaces a valid dual bound (the earlier smoke-test table predates this capability), so global optimality can be certified on many imported MINLPLib instances; the remaining gap is concentrated in high-degree factorable structure, where the lifted relaxation can be loose or expensive to build. The JAX Metal backend (Apple GPU) is currently broken in JAX 0.8.2, so the optional differentiable and learned-relaxation subsystems are CPU-only on macOS; the default solve path does not use JAX and is unaffected. More broadly, the retirement documented in Section 5 has a cost worth naming: discopt no longer has an end-to-end traceable solve. Differentiability is now supplied at the solution point by implicit differentiation rather than by differentiating through the iterations, which covers decision-focused learning but forecloses, say, unrolling the barrier trajectory.

Comparison with existing solvers. discopt is not intended to replace BARON or SCIP on their core strength — solving large-scale MINLP to global optimality via decades-tuned sparse algorithms. The head-to-head of Section 17 is candid about the standing: on *coverage* of a hard $\approx$ 200-instance MINLPLib panel under a fixed 60~s budget, BARON and SCIP each certify on the order of 180 instances to discopt's $\approx$ 120, and discopt's shortfall is a per-node wall-clock / gap-closure gap on a serial engine rather than an unsound bound. What discopt establishes instead is (i) *soundness parity* — zero correctness violations against both mature baselines across the whole panel — and (ii) a complementary capability set absent from monolithic solvers: differentiability, batch node evaluation, memory safety, and Python-native modeling. Closing the coverage gap is active performance work (issues #727, #723, #282); the two-backend seam lets a user cross-check a suspect solve against Ipopt without changing a line of the model.

Command-line interface and literature monitoring. discopt provides a command-line interface (`discopt`) installable via `pip`, with subcommands for searching arXiv and OpenAlex APIs and writing structured reports. These subcommands output JSON, enabling integration with scripting workflows and CI pipelines. A companion "discoptbot" agent uses the CLI to periodically scan for new publications relevant to the solver's algorithmic components (spatial B&B, McCormick relaxations, GNN branching, GPU-accelerated IPM), score them by relevance, and produce structured literature reports. This automated monitoring helps maintain awareness of algorithmic advances that could inform future solver development.

Future work. Key directions include: (1) GPU-accelerated batch node evaluation, which the retirement of the JAX IPM leaves as an open question rather than a solved one —

the batching interface survives in POUNCE, but a Rust backend does not inherit an XLA GPU path for free; (2) extending POUNCE with sparse linear algebra to compete with Ipopt on large problems, and addressing the per-solve POUNCE-versus-Ipopt gap localised in the performance re-profile; (3) learned relaxations, revisited only with a soundness certificate and a measured bound gain, since the experimental ICNN route was retired after showing neither; (4) meta-learning of solver parameters through the differentiable solver, and extending the bilevel module (Section 4.9) to integer and nonconvex lower levels beyond the current convex-KKT reduction; (5) extending the DAE collocation module to higher-index DAEs, adaptive mesh refinement, and multi-stage optimal control with path constraints; (6) extending the neural network embedding to convolutional layers, residual connections, gradient-boosted tree models, and LP-based bound tightening (OBBT) for tighter big-M constants; and (7) extending the uncertainty stack (Sections 4, 4.8) beyond the current affine decision rules and two-stage recourse — to bilinear coefficient uncertainty, polynomial or piecewise-linear recourse policies, multistage (nested-Benders / SDDP) recourse, and a full single-level distributionally-robust reformulation over moment-based or Wasserstein ambiguity sets.[79,81]

19 Conclusion

We have presented discopt, a hybrid Rust/Python MINLP solver that bridges the gap between classical mathematical programming and modern differentiable programming. The solver provides two interchangeable NLP backends—POUNCE (a memory-safe Rust translation of Ipopt, developed as a separate companion project) and Ipopt via cyipopt—producing identical results on benchmark problems. A third, a pure-JAX interior point method, was the project's original solver and was retired once POUNCE matched Ipopt's speed: the honest summary of that experiment is that building the numerical core in an autodiff framework was not what made the solver differentiable, and the framework's cost was paid on every solve while its benefit was needed on few. The default path is now Rust and `numpy` end to end, and JAX is an optional dependency of the differentiable-solving and sensitivity subsystems rather than a layer of the solver. The differentiable solving capability enables decision-focused learning through nonconvex optimization, achieving 36% regret reduction on resource allocation tasks. Piecewise McCormick with $k = 8$ uniform partitions, the relaxation behind AMP, reduces the mean univariate envelope gap by 97–98%, and the algebraic coefficient extraction path provides 17–226× speedups for structured subproblems. The neural network embedding module enables optimization over trained ML surrogates by formulating feedforward networks as MINLP constraints with big-M, full-space, and reduced-space strategies. POUNCE eliminates the Fortran/C++ dependency chain while preserving Ipopt's numerical robustness.

The system is open-source: discopt is available under the EPL-2.0 License, and POUNCE is available as a separate companion project (`https://github.com/jkitchin/pounce`) under the EPL-2.0 License.

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